How Will Paid AI Spread? Evidence from Home Computers in US Households, 1984–2013

A test of the profit switch hypothesis with US household surveys and computer price indexes

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Abstract

The spread of home computers through US households is a guide to how paid AI may spread. Every household faced the same falling computer prices, but a price cut does most for households close to affording a computer, so *poorer households gained from falling prices last.* The poorest fifth of US households, ranked by income, matched the richest fifth's 1996 computer ownership rate, about 60%, only in 2012. We study 178,891 households in the Consumer Expenditure Survey (January 1994 to March 2013), 369,393 households in eight Current Population Survey supplements (October 1984 to October 2003), and the Bureau of Economic Analysis computer price index. We find three key results. First, *falling quality-adjusted computer prices, far more than rising incomes,* paced the rise in the share of US households that owned a computer, from 24% in 1994 to 81% in 2012, and accounted for 53 of its 57 percentage points. Second, the gap in computer ownership between the richest and poorest fifths of households *widened, then narrowed*, from 39 percentage points in 1994 to 54 in 2003 and 37 in 2012. Third, falling prices *left the education gap in computer ownership in place.* Averaged over 1994–2012, a household headed by a college graduate was as likely to own a computer as a household headed by a high-school graduate that spent 2.8 times as much. If paid AI follows this pattern, paid AI may reach lower-income households *more than a decade after richer households*, unless policy subsidizes their access and helps them learn to use it.

Summary of core findings

Every household faced the same falling computer prices, but poorer households gained from falling prices last. The poorest fifth of US households, ranked by income, matched the richest fifth's 1996 computer ownership rate, about 60%, only in 2012, 16 years later. Falling quality-adjusted computer prices, far more than rising incomes, paced the spread of home computers but left the education gap in computer ownership in place. If paid AI spreads the way home computers did, paid AI may reach lower-income households more than a decade after richer households, unless policy subsidizes their access to paid AI and helps them learn to use it.

Once a technology reaches 20–30% of households, affordability (what households can spend relative to its price) should pace its further spread. We test this hypothesis on the home computer with 178,891 households interviewed in the Consumer Expenditure Survey from January 1994 to March 2013, the Bureau of Economic Analysis price index for personal computers, and 369,393 households in eight Current Population Survey supplements from October 1984 to October 2003. Our three key results cover 1994 to 2012, the last full year of Consumer Expenditure Survey data.

1. **Falling quality-adjusted computer prices, far more than rising incomes, paced the spread of computers.** Of the rise in the share of US households that owned a computer, from 24% in 1994 to 81% in 2012, our model attributes 53 percentage points to the falling quality-adjusted computer price (the price of a computer of fixed capability) and 3 points to rising household spending after inflation. Across the 20 years from 1994 to 2013, the spending threshold, the yearly spending at which a household was more likely than not to own a computer, fell by 4–6% for each 10% fall in the quality-adjusted price, only about half the 10% that affordability alone predicts. But from 1994 to 2012 the quality-adjusted price fell 98%, while the dollar price that households paid for a computer fell by only 7% at the median, because the gains in technology went into better computers and into *productization*, the work of turning the computer into a product that ordinary households could buy and use.
2. **As computer prices fell, the gap in computer ownership between rich and poor households widened and then narrowed.** The share of households that owned a computer was 39 percentage points higher in the richest fifth of households, ranked by income, than in the poorest fifth in 1994, 54 points higher in 2003, and 37 points higher in 2012. A price cut is most likely to tip households close to affording a computer into buying one, and the households close to affording one were the richest households in 1994 and poorer households after 2003, when most rich households owned one.
3. **Falling computer prices over time still left the education gap in computer ownership in place.** In 1994, 45% of households headed by a college graduate owned a computer, against 15% of households headed by a high-school graduate. This gap of 30 percentage points in computer ownership grew to 39 points in 1997 and narrowed only after 2001, once more than 80% of college households owned a computer. Averaged over 1994–2012, a household headed by a college graduate that spent $10,000 a year was as likely to own a computer as an otherwise similar household headed by a high-school graduate that spent $28,000, and this multiple of 2.8 did not shrink as computer prices fell.

Policymakers will also need a quality-adjusted price index for AI services to track how affordable AI is becoming. Once most households pay for AI, a given income gain would raise paid AI use most if it went to the bottom half of households, but cheaper AI alone may leave more educated households ahead.

1 Introduction

Home computers spread to most US households between 1984 and 2012 as computers became far cheaper for what they could do, and *poorer households gained from falling computer prices last.* In 1984, 8% of US households owned a computer, according to the Current Population Survey, and by 2012, 81% did, according to the Consumer Expenditure Survey. Every household faced the same computer prices, yet the poorest fifth of households, ranked by income, matched the richest fifth's 1996 computer ownership rate, about 60%, only in 2012, 16 years later. From 1994 to 2012 alone, the *quality-adjusted price* of a personal computer, which is the price of a computer of fixed capability, fell to 1/56 of its 1994 level, while the median household's spending rose by 61% in current dollars and by 14% after inflation. This paper asks which of these two changes, falling computer prices or rising household incomes, paced the spread of home computers, how the two worked together, and why poorer households gained from falling prices last.

The answer bears on how paid artificial intelligence (AI) services may spread through households. Generative AI has spread faster than the personal computer did. In August 2025, less than three years after the release of ChatGPT, 54.6% of US adults aged 18 to 64 used generative AI. In 1984, three years after the IBM PC went on sale, 19.7% of US adults aged 18 to 64 used a personal computer (Bick et al., 2025). Households pay nothing in dollars for much of the use of generative AI. If paid AI services follow the pattern of home computers, the history of the home computer can show which households may take up paid AI as its price falls, and when. If paid AI spreads the way home computers did, paid AI may reach lower-income households more than a decade after richer households, unless policy subsidizes their access to paid AI and helps them learn to use it.

The answer also bears on the *profit switch hypothesis*, which holds that what paces the spread of a new technology in the country that invents it changes as the technology spreads. Until about 20–30% of households have adopted the technology, firms' profit share, the share of national income that goes to profits, paces the spread through the investment that lowers the cost of making the technology. After that point, *affordability*, meaning what households can spend relative to the price of the technology, paces the spread. The personal computer was developed and first sold in volume in the United States, and US household surveys record computer ownership from 1984, so US home computers make a natural first test of the hypothesis.

In this paper, we use two US household surveys that together cover 1984 to 2013. The Consumer Expenditure Survey (CE) of the Bureau of Labor Statistics records computer ownership and spending for 178,891 households interviewed from January 1994 to March 2013, and all of our main findings come from the CE. Because the 2013 CE files cover only January to March, we measure changes from 1994 to 2012, years in which the share of US households that owned a computer, which we call the *computer ownership rate*, rose from 24% to 81%. The Current Population Survey (CPS) records computer ownership and family income for 369,393 households in eight supplements from October 1984 to October 2003, and only the CPS covers the years before the computer ownership rate reached 20%. We measure each household's income by its *long-run spending*, the household's average spending across its survey interviews at an annual rate, which predicts computer ownership better than reported income does in every year from 1994 to 2012.

We test affordability by asking whether the spending a household needed before it owned a computer fell in step with the quality-adjusted computer price. For each year from 1994 to 2013 we estimate the *spending threshold*, the level of long-run spending at which a household with average location and traits has a 50% chance of owning a computer, and we then regress the 20 yearly spending thresholds on the quality-adjusted computer price. Under *pure affordability*, the computer price

and household spending are all that matter for whether a household owns a computer. When the computer price halves, the spending a household needs before it buys a computer also halves, so in a regression of the log spending threshold on the log computer price and a time trend, the coefficient on the log computer price is 1 and the coefficient on the time trend is 0 (Section 2). We call the first coefficient, b, the *price effect* and the second, d, the *unexplained trend*.

We estimate one baseline regression and extend it in two directions. The baseline regression uses observed long-run spending and the household's state or region. The extended regression corrects long-run spending for noise between interviews and adds household traits and the education of the household head, and the CPS extension reaches back to 1984 to test whether the unexplained trend changed when the computer ownership rate passed 20–30%.

Our main finding is that from 1994 to 2012 falling quality-adjusted computer prices, far more than rising household incomes, paced the spread of home computers, first among rich households and then among poorer ones, but left the education gap in computer ownership in place.

First, **falling computer prices did almost all the work of spreading home computers from 1994 to 2012.** In constant dollars, our model attributes 53 percentage points of the 57-point rise in the computer ownership rate from 1994 to 2012 (94% of the part that prices and incomes explain) to the falling computer price and 3 points (6%) to rising household spending. Falling prices did this much even though the spending threshold followed the quality-adjusted computer price at *only about half the strength* that pure affordability predicts (Figure 2). Our estimates imply that, across the 20 years from 1994 to 2013, a 10% fall in the quality-adjusted computer price lowered the spending threshold by 4.6% in the baseline regression ($b = 0.45$) and by 6.0% in the extended regression ($b = 0.59$). Both 90% intervals for b exclude 1, the value that pure affordability predicts, while the estimated unexplained trend lies within half a percent a year of zero, as pure affordability also predicts. Half strength was enough because from 1994 to 2012 the computer price relative to all consumer prices fell to 1/79 of its 1994 level, while median long-run spending rose by only 14% after inflation.

The price that fell was the quality-adjusted price, which fell because each dollar bought a better computer. From 1994 to 2012 the dollar price that households paid for a computer fell by 7% at the median and by about half at the 90th percentile, while the quality-adjusted computer price fell 98%. Better computers account for 84–98% of the 1994–2012 fall in the quality-adjusted price, measured in logs, and lower dollar prices account for the rest. The dollar price fell far less because the gains in technology went into better computers and into *productization*, the work of turning the computer into a product that ordinary households could buy and use. Measured against the dollar price, affordability explains *almost none* of the spread of computers from 1994 to 2012.

Second, as computer prices fell, **the gap in computer ownership between rich and poor households widened and then narrowed**. In the raw survey data, the share of households that owned a computer was 39 percentage points higher in the richest fifth of households, ranked by reported income, than in the poorest fifth in 1994, 54 points higher in 2003, and 37 points higher in 2012, slightly less than in 1994. Every household faced the same computer price, but in the household model (Figure 6) a price cut is most likely to tip households whose spending is near the spending threshold into buying a computer. In 1994 the spending threshold was above the spending of 95% of households, so the households close to buying a computer were among the *highest* spenders. By 2012 the spending threshold was below the spending of 96% of households, so the households close to buying were among the *lowest* spenders.

Third, **falling computer prices still left the education gap in computer ownership in**

place. In the raw survey data, 45% of households headed by a college graduate owned a computer in 1994, against 15% of households headed by a high-school graduate, and this gap of 30 percentage points in computer ownership grew to 39 points in 1997 and narrowed only after 2001, once more than 80% of college households owned a computer. Pure affordability rules out any gap in computer ownership between education groups among households with the same spending, yet averaged over 1994–2012, a household headed by a college graduate that spent $10,000 a year was as likely to own a computer as an otherwise similar household headed by a high-school graduate that spent $28,000, and this multiple of 2.8 *did not shrink* as computer prices fell (Figure 7).

These estimates hold up in robustness checks. Fitting the extended regression's two stages as one Bayesian model of all 178,891 CE households of 1994–2013 gives $b = 0.60$, against 0.59 from the two stages, and most other choices of sample, spending measure, and link function leave b between 0.37 and 0.61 (Section 5.4).

Our approach builds on threshold models of the spread of durable goods, such as cars and refrigerators, in which a falling threshold passes down through the income distribution (David, 1969; Bonus, 1973; Geroski, 2000), and on Griliches (1957), who showed that the profitability of hybrid corn explains how fast and how far hybrid corn spread across US states. We estimate the spending threshold directly in each year and test its path against an explicit benchmark, $b = 1$ and $d = 0$, that the data can reject. Greenwood et al. (2005) argue that technical progress lowered the prices of household appliances and spurred their adoption, and they note that poor households bought durable goods later than rich households, when prices were lower. We measure how much of the spread of home computers came from falling prices and how much from rising incomes, and we show how the households that a price cut helps most, those close to affording a computer, moved down the income distribution. Hedonic price indexes adjust the price of a computer for its quality (Berndt and Rappaport, 2001; Pakes, 2003), and measures based on computing performance show larger price falls than the official indexes (Nordhaus, 2007). We show that from 1994 to 2012 the spread of home computers followed the quality-adjusted computer price, at about half strength, while the dollar price that households paid barely fell.

Our estimates also bear on earlier work about how income, education, and spillovers among computer users shape who owns a computer. Chinn and Fairlie (2007) find that differences in income account for most of the gap in computer and internet use across countries. Nelson and Phelps (1966) argued that education speeds the adoption of new technology, and Riddell and Song (2017) find that more schooling raises the chance that a worker uses a computer on the job. Goolsbee and Klenow (2002) find that US households were more likely to buy a first home computer in areas where more households already owned one, which is evidence of spillovers from learning or from networks of computer users. Spillovers that grew with national computer ownership would show up in our regressions as a negative unexplained trend, and we find no such trend in the national series for 1994 to 2013, although 20 yearly estimates cannot rule out small trends. The paper also gives a household-level test of the affordability phase of the profit switch hypothesis, which companion work studies with cross-country data on internet adoption.

The estimates show how closely the spending threshold followed the computer price, but the data limit what the estimates can say about causes and about the early years of the spread of computers. The computer price is the same for every household in a given year, so b and d rest on the 20 yearly spending thresholds of 1994–2013, and they also pick up anything else that moved with computer prices over time, such as the spread of the internet and of software. The household data can measure how much of the fall in the spending threshold the computer price leaves unexplained, but the household data cannot say what caused that unexplained part. The surveys also cannot

test whether the forces behind computer ownership changed at 20–30% ownership, because only two surveys, the CPS of 1984 and 1989, measure computer ownership below 20% (Appendix B.3).

If paid AI services follow the pattern that home computers followed after 1994 (Section 7), policymakers will need a quality-adjusted price index for AI services to track how affordable AI is becoming, because the subscription price, like the dollar price of a computer, may fall far less than the quality-adjusted price. Every household would face the same price for an AI service, but a cut in that price would be most likely to tip households close to affording paid AI into paying for it, so the gap between the shares of rich and poor households that pay for AI may widen before it narrows. Unless policy subsidizes lower-income households' access to paid AI, paid AI may reach those households more than a decade after richer households, as home computers did. Once most households pay for AI, a given income gain would raise paid AI use most if it went to the bottom half of households, but cheaper AI alone may leave more educated households ahead, so help in learning to use AI may matter as well as its price. These implications hold *only for AI that households pay for in dollars*. Firms can also earn from free AI indirectly, through advertising, which OpenAI said in January 2026 it would test in the free tier of ChatGPT (OpenAI, 2026), and through the conversations that users allow firms to use to train their models (Anthropic, 2025). How far AI funded in these ways spreads may depend more on what firms earn from it than on what households can spend. To test the profit switch hypothesis for AI, statistical agencies would also need a household survey that records AI use, paid or free, and income every year from early adoption on.

Section 2 sets out a household model of computer ownership and derives what pure affordability predicts. Sections 3 and 4 describe the data and the method, Section 5 presents the findings, Section 6 discusses them, and Section 7 draws out the implications for AI diffusion and policy. Appendix A reports the robustness checks, and Appendix B gives further results. The online appendix, which describes how we built the dataset and gives the method and the remaining estimates in detail, is at `https://github.com/IlanStrauss/home-computer-diffusion` with the code and the processed data.

2 A household model of computer ownership

This section sets out when a household owns a computer, derives what pure affordability predicts, and shows how the computer price and household income interact. Section 4 turns the household model into the regressions we estimate.

2.1 When a household owns a computer

Household i owns a computer in year t if the household's long-run spending, relative to the cost of owning a computer, exceeds a cutoff:

$$\ln s_{it} - \ln K_t > c_i, \qquad c_i \sim \text{Normal}(c, \sigma^2), \tag{1}$$

where s_{it} is the household's long-run spending in current dollars a year and K_t is the cost of owning a computer in year t, in current dollars. The cutoff c_i is the amount, in logs, by which the household's spending must exceed the cost of owning a computer before the household buys a computer. The mean cutoff across households is c, and the standard deviation of cutoffs across households is σ, also in logs. A household that values a computer highly has a low cutoff. We write

$x_{it} = \ln s_{it}$ for the household's log long-run spending. The probability that a household with log spending x_{it} owns a computer is

$$\Pr(\text{own}_{it} = 1 \mid x_{it}) = \Phi\left(\frac{x_{it} - \tau_t}{\sigma}\right), \qquad \tau_t = c + \ln K_t, \tag{2}$$

where Φ is the standard normal distribution function. The spending threshold τ_t is the log long-run spending at which the probability of owning a computer is one half. A household whose spending is above the spending threshold is more likely than not to own a computer, and a household whose spending is below the spending threshold is less likely than not to own a computer. The computer ownership rate, the share of households that own a computer, is

$$P_t = \int \Phi\left(\frac{x - \tau_t}{\sigma}\right) dF_t(x), \tag{3}$$

where F_t is the distribution of log long-run spending across households in year t. Equations (1)–(3) are the threshold model of durable-good ownership (Bonus, 1973) and the probit model of diffusion, the spread of a technology across users (David, 1969; Geroski, 2000). The computer ownership rate rises when the spending threshold falls through the distribution of household spending or when the distribution of household spending moves up past the spending threshold.

2.2 What pure affordability predicts

The cost of owning a computer combines the quality-adjusted computer price p_t with other costs k_t, such as software, an internet connection, and the time it takes to learn to use a computer:

$$\ln K_t = \psi \ln p_t + (1 - \psi) \ln k_t, \tag{4}$$

where ψ, between 0 and 1, is the weight of the computer price in the cost of owning a computer. We write $\ell_t = \ln p_t - \ln p_{1994}$ for the log computer price relative to 1994. Suppose that the log of other costs changes by a constant κ a year and that the mean cutoff changes by a constant ν a year. Then equations (2) and (4) give a spending threshold that follows

$$\tau_t = a + b\,\ell_t + d\,(t - 1994), \tag{5}$$

where $a = \tau_{1994}$ is the 1994 spending threshold, $b = \psi$ is the price effect, meaning the change in the log spending threshold per unit change in the log computer price, and $d = (1 - \psi)\kappa + \nu$ is the unexplained trend, meaning the yearly change in the log spending threshold that the computer price does not explain. Because both the spending threshold and the computer price are in logs, b is an elasticity, so a 10% fall in the computer price lowers the spending threshold by about $b \times 10\%$.

Pure affordability means that the computer price is the whole cost of owning a computer ($\psi = 1$) and that the cutoffs c_i of equation (1), the margins by which a household's log spending must exceed the log cost of owning a computer before the household buys one, do not change over time (the mean cutoff c and the spread of cutoffs σ stay fixed, so $\nu = 0$). Equation (2) then gives

$$\tau_t - \tau_{1994} = (c + \ln p_t) - (c + \ln p_{1994}) = \ell_t, \tag{6}$$

which is equation (5) with $b = 1$ and $d = 0$.

Under pure affordability the spending threshold moves in proportion to the computer price. Suppose a household buys a computer once its yearly spending reaches 20 times the computer price. At a

computer price of $2,000 the household must spend $40,000 a year before it buys a computer, and at a computer price of $1,000 the household must spend $20,000 a year. Halving the computer price halves the spending threshold, so in logs the spending threshold moves one-for-one with the computer price ($b = 1$). Nothing else moves the spending threshold, so the spending threshold has no unexplained trend ($d = 0$).

A price effect b below 1 means that the spending threshold fell by a smaller percentage than the computer price. In the household model, a price effect below 1 arises when other costs of owning a computer fell more slowly than the computer price ($\psi < 1$). The price effect also falls below 1 if households valued less of the improvement in computer quality than the price index counts, so that the effective price households faced fell more slowly than the price index. A negative unexplained trend d means that the spending threshold fell faster than the computer price explains, as the spending threshold would if computers grew more useful as more households owned computers, or as households learned about computers (Goolsbee and Klenow, 2002). A positive d means that the spending threshold fell more slowly than the computer price explains.

2.3 How the computer price and household income interact

In the household model, a fall in the computer price and a rise in household spending raise the computer ownership rate through the same channel, the number of households whose spending is close to the spending threshold. Suppose that the log computer price changes by $\Delta\ell$ (negative for a fall) and that every household's log spending rises by g. For small changes, equations (3) and (5) give

$$\Delta P_t \approx D_t\,(g - b\,\Delta\ell), \qquad D_t = \frac{1}{\sigma}\int \phi\!\left(\frac{x-\tau_t}{\sigma}\right) dF_t(x), \tag{7}$$

where ϕ is the standard normal density. D_t measures how many households have spending close to the spending threshold. D_t is the density of log spending around the spending threshold, averaged over a window whose width is set by the spread of cutoffs σ. Equation (7) has three consequences.

1. The computer price and household spending work through the same term. A rise of 1% in every household's spending does as much for the computer ownership rate as a fall of about $1/b$ percent in the computer price, at every level of income.

2. A household's response to the computer price depends on the household's income. The household's probability of owning a computer changes by $-(b/\sigma)\,\phi((x_{it} - \tau_t)/\sigma)$ per unit change in the log computer price. The response is largest for households whose spending equals the spending threshold, and the response is smaller the further a household's spending lies above or below the spending threshold. As the spending threshold falls, the households that respond most to the computer price move down the income distribution.

3. The response of the computer ownership rate to the computer price depends on the distribution of income, through D_t. A fall in the computer price raises the computer ownership rate most when many households have spending near the spending threshold. Once the spending threshold falls below the spending of the median household, D_t depends on the incomes of the bottom half of households. In the same way, the effect of income growth on the computer ownership rate depends on the computer price, which sets where the spending threshold sits.

Online Appendix E.4 tests whether the computer price and household income interact in any other way.

2.4 Predictions

Pure affordability makes three predictions, which the baseline and extended regressions test with the CE households of 1994–2013. Each prediction pairs a hypothesis with the coefficient that measures it and with the value of that coefficient that the hypothesis implies.

1. *Hypothesis:* the spending threshold moves in proportion to the quality-adjusted computer price, and nothing else moves the spending threshold. *Coefficients:* the price effect b and the unexplained trend d of equation (5). *Prediction:* $b = 1$ and $d = 0$, so the 90% interval for b should include 1 and the 90% interval for d should include 0.

2. *Hypothesis:* the spread of cutoffs across households, σ, stays the same from year to year. *Coefficient:* the spending slope $\beta_t = 1/\sigma$, which measures how steeply a household's probability of owning a computer rises with the household's log long-run spending in year t. *Prediction:* β_t is the same in every year from 1994 to 2013 and does not change as the computer price falls.

3. *Hypothesis:* once long-run spending is measured well, the education of the household head does not change the household's cutoff. *Coefficient:* the education shift θ_e, the shift in the log spending threshold for households whose head has education level e, relative to households headed by a high-school graduate (Section 4.1). *Prediction:* $\theta_e = 0$ for every education level, so at the same long-run spending, households with more education are no more likely to own a computer.

The profit switch hypothesis adds a prediction for the years around the switch. Until the computer ownership rate passes 20–30%, firms' profits and investment, which the household model leaves out, pace the spread of computers, and after that point affordability does. The unexplained trends d_1 before and d_2 after the date at which the computer ownership rate passed 20% or 30% (Online Appendix D.4) should then differ, with $b = 1$ and $d_2 = 0$ after the switch. The CPS extension tests this prediction with the CPS supplements of 1984–2003 and the CE (Appendix B.3).

3 Data

3.1 Two household surveys

We use two US household surveys, and Table 1 gives the years, the units, and the number of households of each survey. The main dataset is the Interview Survey of the Consumer Expenditure Survey (CE), a quarterly survey that the Bureau of Labor Statistics (BLS) runs. We use the CE public-use files from the first quarter of 1994 to the first quarter of 2013. The CE interviews each sampled household every three months for five consecutive quarters, and the public files hold the last four of those five interviews, each with the household's spending in the three months before the interview. The CE calls a household a consumer unit, which is a household or a financially independent group within a household, and we call consumer units households.

The CE files for 1994 to 2013 hold 522,833 interviews with 178,920 households that answered the appliance inventory, the part of the interview in which the household lists its major appliances. Our analysis sample has 178,891 of those households, one record per household: 6,153 to 13,743 households in each year from 1994 to 2012, and 2,203 in 2013, the year whose files cover only the

first quarter. The first stage fits one regression for each year, so each first-stage regression uses 2,203 to 13,743 households. The second stage regresses the 20 yearly spending thresholds, one for each year from 1994 to 2013, on the computer price, so the second stage has 20 observations. The joint model of the robustness checks fits all 178,891 households at once. With survey weights, the CE computer ownership rate was 23.8% in 1994, 64.2% in 2003, and 80.5% in 2012. Every estimate in Section 5 comes from the CE. Because the 2013 files cover only January to March, changes over time in this paper run from 1994 to 2012, the last full year of CE data, while the second stage uses all 20 years from 1994 to 2013.

The second dataset is the eight Computer Use supplements to the Current Population Survey (CPS), the monthly household survey that the Census Bureau runs for the BLS. The supplements ran in October 1984, 1989, 1993, and 1997, December 1998, August 2000, September 2001, and October 2003, and each supplement asks whether the household has a computer. The CPS records family income in brackets and does not record spending. The CPS extension uses 369,393 CPS households from October 1984 to October 2003, 40,011 to 51,162 in each supplement. The CPS gives household computer ownership rates of 8.2% in October 1984, 15.0% in October 1989, and 61.8% in October 2003, within 0.3 percentage points of the Census Bureau's published figures. The CPS is the only one of the two surveys that covers the years before the computer ownership rate reached 20% (Appendix B.3).

We keep one record per CE household, from the household's first public interview with the appliance inventory, and the household belongs to the year of that interview. Long-run spending, which leaves out vehicle purchases, spending on computers and internet service, personal insurance and pensions, and cash contributions, is four times the household's average spending per three-month interview period. The CE takes the appliance inventory at the household's first interview, which is not in the public files, so we date most inventories three months before the household's first public interview, and each household gets the computer price for that *inventory month*. Online Appendix C.1 lists everything we do to prepare the two surveys, including the records we drop.

3.2 Computer prices

The main computer price is the Bureau of Economic Analysis (BEA) price index for personal consumption expenditures (PCE) on personal computers/tablets and peripheral equipment (National Income and Product Accounts, NIPA, Table 2.4.4U, line 49). This index is quality-adjusted, so when a new computer costs the same as last year's model but is faster and holds more data, the index records a price fall. From 1994 to 2012 the log main computer price falls by 4.02, so the main computer price falls to 1/56 of its 1994 level. We also use the dollar price that CE households reported paying for computers, as a weighted median and 90th percentile (Section 5.1), and, as checks, three other quality-adjusted price indexes (Appendix A.4). Household spending and computer prices are in current dollars unless the text says constant dollars.

3.3 What the raw data show

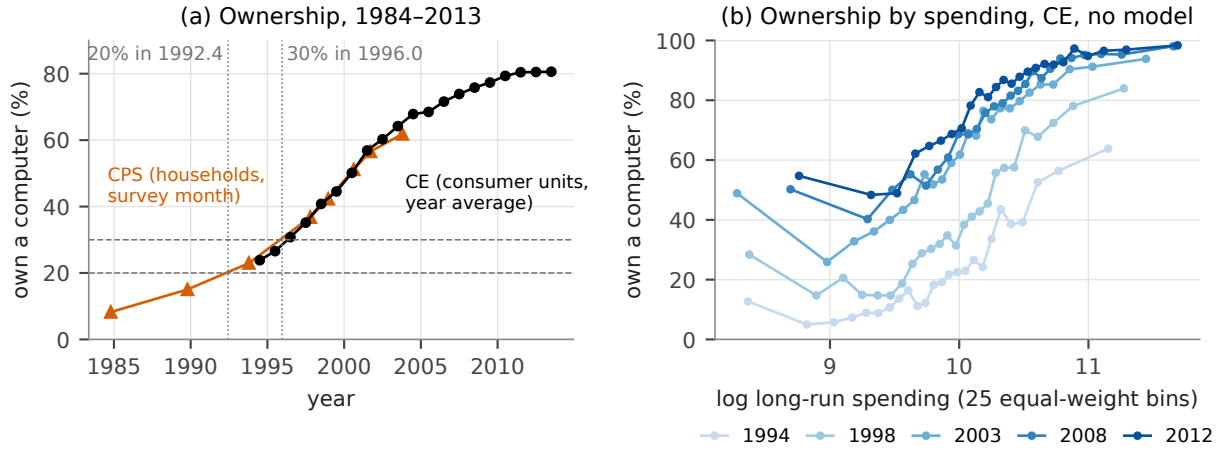
The raw data show the computer ownership rate rising along an S-shaped path from 1984 to 2013, and the curve that relates computer ownership to household spending moving toward lower spending as the years passed. Figure 1a shows the computer ownership rate of US households in the eight CPS supplements from October 1984 to October 2003 and in each CE year from 1994 to 2013. In the five years that both surveys cover (1997, 1998, 2000, 2001, and 2003), the CPS

Table 1. The two datasets, the CE (January 1994 to March 2013) and the CPS supplements (October 1984 to October 2003). Households in the analysis are the households in the fits, which for the CE means one record per household (Online Appendix C.1) and for the CPS means households with a known income bracket and known traits.

	CE Interview Survey	CPS Computer Use supplements
Years and frequency	Every quarter, from the first quarter of 1994 to the first quarter of 2013	Eight supplements: October 1984, 1989, 1993, and 1997; December 1998; August 2000; September 2001; and October 2003
Unit	Household (the CE's consumer unit)	Household
Households in the analysis	178,891: 6,153 to 13,743 a year from 1994 to 2012, and 2,203 in 2013	369,393: 40,011 to 51,162 a supplement
Observations in each regression	First stage: one probit model for each year, with 2,203 to 13,743 households. Second stage: the 20 yearly spending thresholds of 1994–2013. Joint model: all 178,891 households	First stage: one probit model for each supplement, with 40,011 to 51,162 households. Switch test: 28 spending thresholds, 8 from the CPS and 20 from the CE
Computer ownership	The appliance inventory lists "computers, not solely for games" (to the first quarter of 2001) or a "home computer" (from the second quarter of 2001)	"Is there a computer in this household?", with small changes of wording from 1998
Income	Long-run spending, from all of the household's interviews	Family income in brackets, from which we impute long-run spending
Household traits	State or region, age of the household head, numbers of children and adults, marriage, race, Hispanic origin, student housing, and education	The same, except student housing
Weights	Survey weights and 44 replicate weights	Household weights, with no replicate weights
Used in	Every regression	The CPS extension

and CE computer ownership rates lie within 2.4 percentage points of each other. In the CPS the computer ownership rate rose slowly from 1984 to 1993, from 8.2% to 23.0%. In the CE it rose fastest from 1996 to 2001, by 3.7 to 6.7 percentage points a year, and more slowly from 2004 on, by 0.1 to 3.1 points a year, to 80.5% in 2012. Figure 1b shows, with no model, the share of CE households that owned a computer at each level of long-run spending in 1994, 1998, 2003, 2008, and 2012. In each of these five years the share of households owning a computer rises in an S-shape with log long-run spending, apart from the lowest spending bins, where students raise computer ownership (Online Appendix C.5). The curve for each later year lies to the left of the curves for earlier years, with little change in its steepness. The threshold model of Section 2 assumes such a shift, with one S-shaped curve for each year that moves down the spending axis as the years pass.

Figure 1. The share of US households that owned a computer rose along an S-shaped path from 1984 to 2013, while the spending at which households owned a computer fell



(a) Share of households that own a computer: CPS households in each of the eight survey months from October 1984 to October 2003, and CE households as a yearly average from 1994 to 2013, with survey weights. Dotted lines mark when the computer ownership rate passed 20% and 30% (log-odds interpolation between CPS surveys). (b) Share of CE households that own a computer against log long-run spending, in 25 bins of equal weight, for 1994, 1998, 2003, 2008, and 2012, with no model. A log spending of 10 is $22,000 a year and a log spending of 9 is $8,100.

4 Method

The method has two stages. The first stage fits a probit model separately for each year from 1994 to 2013, to that year's CE households alone, and estimates the year's spending threshold. The second stage regresses the 20 yearly spending thresholds on the computer price. Estimating a separate model for each year, and only then modelling how the yearly estimates change, lets the data show the path of the spending threshold before we impose any model of that path.

The subscripts run as follows. Years are $t = 1994, \ldots, 2013$, which makes 20 years. Households in year t are $i = 1, \ldots, N_t$, where N_t is the number of CE households in year t: 6,153 to 13,743 in each year from 1994 to 2012 and 2,203 in 2013, so that $N = \sum_t N_t = 178{,}891$ households in all. Geography groups are $g = 1, \ldots, 29$. The household-trait vector z_{it} has K entries. $K = 0$ in the baseline regression and with corrected spending alone, $K = 16$ once household traits enter, and $K = 19$ in the extended regression, which adds three education levels of the household head, $e \in \{$less than high school, some college, college degree$\}$, each measured against high school. Table 2 lists the regressions. We wrote these regressions and the tests of pure affordability into an analysis plan before we ran any of them, and Online Appendix H lists where the paper departs from that plan.

4.1 First stage: a spending threshold for each year

For each year t, we fit a probit model, weighted by the CE survey weights, to the year's N_t households. A probit model is a regression model for a yes-or-no outcome, here whether the

Table 2. The regressions and the years of data that each uses. The extended regression adds three things to the baseline regression, one at a time, and Appendix A.1 reports each intermediate fit.

Regression	Data	What the regression does
Baseline	CE, 1994–2013	Estimates a spending threshold for each year from observed long-run spending and the household's state or region, and regresses the 20 yearly spending thresholds on the computer price.
Extended	CE, 1994–2013	Corrects long-run spending for noise between interviews, then adds household traits (age, the numbers of children and adults, marriage, race, Hispanic origin, and student housing), then adds the education of the household head (the CE's reference person, who owns or rents the home).
CPS extension	CPS, 1984–2003, and CE	Estimates spending thresholds in the CPS, with long-run spending imputed from bracketed income, and tests whether the unexplained trend changed when the computer ownership rate passed 20% or 30%.
Robustness checks	CE, 1994–2013	Fits the extended regression's two stages as one Bayesian model, and repeats the regressions with other samples, spending measures, and price indexes.

household owns a computer. The probit model is the empirical form of equation (2):

$$\Pr(\text{own}_{it} = 1) = \Phi\big(\alpha_t + \beta_t(x_{it} - 10) + \gamma_{g(i),t} + z_{it}'\delta_t\big), \qquad \tau_t = 10 - \frac{\alpha_t + \sum_g \pi_g \gamma_{g,t}}{\beta_t}, \tag{8}$$

where

- $\text{own}_{it} = 1$ if household i, observed in year t, has a computer in its appliance inventory;
- x_{it} is the household's log long-run spending (baseline regression) or its log long-run spending corrected for noise between interviews (extended regression), and subtracting 10 (about \$22,000 a year) centres x_{it};
- α_t is the year's intercept and β_t is the year's spending slope;
- $\gamma_{g,t}$ is the effect of geography group g in year t, $g(i)$ is the geography group of household i, and π_g is group g's pooled 1994–2013 share of households, so that moves of households between places do not shift τ_t;
- z_{it} is the vector of the household's K traits, centred at their pooled 1994–2013 means, and δ_t holds the K effects of the traits in year t;
- τ_t is the spending threshold of a household with the pooled average traits in the pooled average geography, the log long-run spending at which this household's probability of owning a computer is one half.

There are 29 geography groups: the 24 states (counting the District of Columbia) that have households in every year and every replicate sample, the rest of each of the four Census regions, and a group for the 1% of households with no state or region, which is fitted but left out of τ_t. The

household traits are the age of the household head (the CE's reference person, the person who owns or rents the home), in seven groups, the numbers of children (none to three or more) and adults (one to three or more), marriage, race (white, black, other), Hispanic origin, and student housing. The extended regression adds the education of the household head in four groups (less than high school, high school, some college, and a college degree), with high school as the reference group. We report an education effect as a shift in the log spending threshold, $\theta_{e,t} = \delta_{e,t}/\beta_t$. A positive θ means that households with that education own a computer at lower spending than households headed by a high-school graduate.

The extended regression corrects long-run spending for noise between interviews by regression calibration (Carroll et al., 2006, ch. 4), which replaces each household's measured log long-run spending with its expected true value (Online Appendix D.1). We compute standard errors by balanced repeated replication. We repeat the whole estimation, including the spending correction, with each of the survey's 44 replicate weights, and the variance of an estimate is the mean squared deviation of its replicate estimates from the full-sample estimate.

We use the probit model in every regression. The analysis plan's rule for choosing the link function, the curve that turns household spending into a probability of owning a computer, would switch two of the four CE regressions to a robit model, whose curve is like the probit curve with heavier tails. Online Appendix D.1 describes the rule, and Online Appendix H shows that the robit model changes the price effect b little.

4.2 Second stage: the spending thresholds and the computer price

We regress the 20 yearly spending thresholds, for $t = 1994, \ldots, 2013$, on the computer price in a two-level Bayesian model (Lewis and Linzer, 2005), the empirical form of equation (5):

$$\hat{\tau}_t \sim \text{Normal}(\tau_t, \text{se}_t^2), \qquad \tau_t = a + b\,\ell_t + d\,(t - 1994) + e_t, \qquad e_t \sim \text{Normal}(0, \omega^2), \tag{9}$$

where $\hat{\tau}_t$ and se_t are the first-stage spending threshold for year t and its standard error, a is the 1994 spending threshold, b is the price effect, d is the unexplained trend (the yearly change in the log spending threshold), and e_t is year t's deviation from the fitted line beyond sampling error, with standard deviation ω. Pure affordability means $b = 1$ and $d = 0$. The model is hierarchical. Its first level lets each year's estimated spending threshold differ from the year's true spending threshold by sampling error, and its second level relates the 20 true spending thresholds to the computer price, so the second stage has 20 observations. Online Appendix D.2 gives the priors, which are weak, and the sampler settings.

Computer prices vary only by year, so b and d rest on the 20 yearly spending thresholds of 1994–2013, and the 178,891 households matter through the precision of those 20 spending thresholds. The second stage can tell b from d only because the computer price fell at different speeds, 27% a year from 1994 to 2003 and 12% a year from 2003 to 2012. If the spending threshold fell at the same fraction of the computer price's speed in both periods, that fraction is b and the unexplained trend is zero. If the spending threshold fell at the same speed in both periods, whatever the speed of the computer price, the unexplained trend d accounts for the fall. Even with this variation in the speed of the computer price, the estimates of b and d are tied. Under the main computer price, the posterior correlation of b and d is 0.96 to 0.98 in every specification. A larger b goes with a larger d, so the data cannot fully separate a stronger price effect plus an upward unexplained trend from a weaker price effect plus a downward unexplained trend.

4.3 Splitting the rise between the computer price and household spending

We split the 1994–2012 rise in the computer ownership rate between the computer price and household spending by changing one at a time in the household model of the baseline regression. Sections 5.1 and 5.2 use the baseline regression's spending thresholds and spending slopes of 1994–2012, its price effect $b = 0.45$, and each year's observed distribution of household spending (`income_price.py`). A household in the pooled average geography owns a computer with probability $\Phi(\hat{\beta}_t(x_{it} - \hat{\tau}_t))$, and the computer ownership rate is the weighted mean of that probability over the year's households. We start from the 1994 spending threshold and spending slope, and we move the computer price and household spending from their 1994 values to their 2012 values one at a time. We then average the contribution of the computer price, and the contribution of household spending, over the two possible orders of change (a Shapley split). We compute the response of the computer ownership rate to a 10% cut in the computer price the same way, as the change in each household's predicted probability of owning a computer, averaged over households, which is an average predictive comparison in the sense of Gelman and Pardoe (2007). The intervals for these quantities allow for the uncertainty in b only.

5 Findings

From 1994 to 2012 falling quality-adjusted computer prices, far more than rising incomes, paced the spread of home computers (Section 5.1), first among rich households and then among poorer ones (Section 5.2), but they left the education gap in computer ownership in place (Section 5.3). This section reports estimates from the 178,891 CE households of 1994–2013 and the main computer price, the BEA price index for personal computers. Throughout, intervals are 90% intervals in square brackets. For a second-stage coefficient the interval runs from the 5th to the 95th percentile of the posterior distribution, and for a yearly spending threshold it is ±1.645 replicate standard errors. Dollar amounts are in current dollars unless the text says constant dollars. Appendix A gives the estimates of every regression in tables, and Table B1 in Appendix B lists the main estimates in plain units.

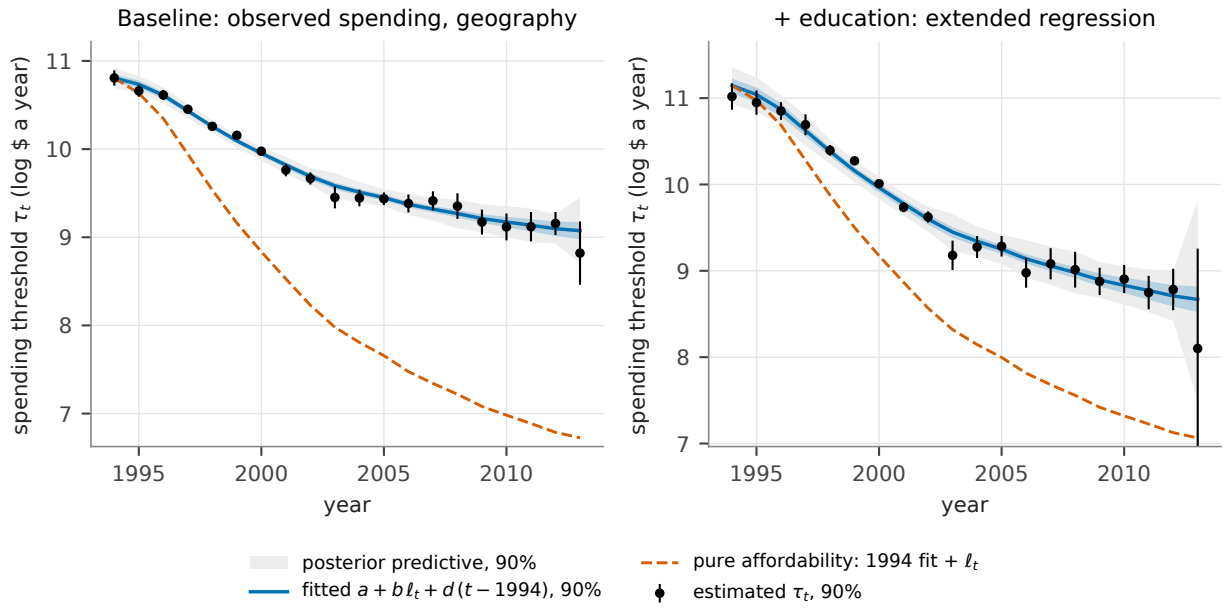
5.1 Falling quality-adjusted computer prices, far more than rising incomes, paced the spread of computers

Falling prices account for 53 of the 57 percentage points by which the computer ownership rate rose from 1994 to 2012, and rising household spending for 3, even though the median price that households paid for a computer fell *only 7%*. The spending threshold followed the quality-adjusted computer price at only about half the strength that pure affordability predicts, but that price fell 98% from 1994 to 2012, because each dollar bought a better computer.

The spending threshold fell fast in the 1990s, when the computer price fell fast, and slowly after 2003, when the computer price fell slowly, but it fell *far less* than pure affordability predicts. Figure 2 plots the spending threshold in each year from 1994 to 2013, in the baseline regression and the extended regression, together with the fitted second stage and the path that pure affordability predicts from the 1994 spending threshold. In the baseline regression the spending threshold fell from $49,400 a year in 1994 to $12,700 in 2003 and $9,500 in 2012. From 1994 to 2012 the spending threshold fell by 81% in the baseline regression and by 89% in the extended regression, while the quality-adjusted computer price fell by 98%. In both regressions the yearly spending thresholds lie

close to the fitted second stage and far above the path that pure affordability predicts.

Figure 2. The spending threshold fell with the quality-adjusted computer price from 1994 to 2013, but far less than pure affordability predicts



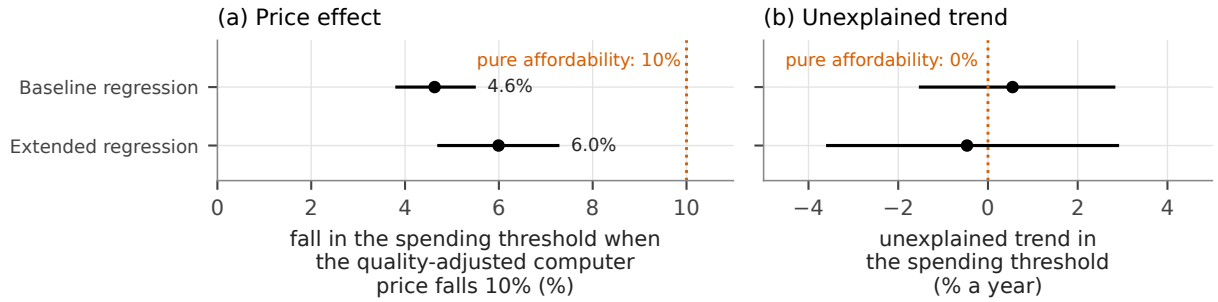
Yearly spending thresholds $\hat{\tau}_t$, 1994–2013, in the baseline regression and the extended regression, with 90% intervals (± 1.645 replicate standard errors). Figure A1 in Appendix A.1 adds the two intermediate fits. The spending threshold is the log long-run spending at which a household with average location and traits has a 50% chance of owning a computer. Blue: the fitted second stage $a + b\,\ell_t + d\,(t - 1994)$ with its 90% interval. Grey: the 90% posterior predictive interval, the range in which the fitted second stage expects each year's estimated spending threshold to fall. Dashed: the path that pure affordability predicts, which is the fitted 1994 spending threshold plus ℓ_t. Main computer price (BEA PCE price index for personal computers). A log spending of 10 is $22,000 a year and a log spending of 9 is $8,100.

Across the 20 years from 1994 to 2013, a 10% fall in the quality-adjusted computer price was associated with a 4–6% fall in the spending threshold, about half the 10% fall that pure affordability predicts. Figure 3 puts the second-stage estimates on a percentage scale. In the baseline regression the price effect is $b = 0.45$ $[0.37, 0.54]$, so a 10% fall in the computer price lowered the spending threshold by 4.6%. In the extended regression the price effect is $b = 0.59$ $[0.46, 0.72]$, so the same fall in the computer price lowered the spending threshold by 6.0%. Both 90% intervals lie below 10% (Figure 3a), and Appendix A.1 shows how each part of the extended regression moves the price effect.

Once the computer price is taken into account, the spending threshold shows no other steady change from 1994 to 2013. The unexplained trend is +0.5% a year $[-1.5\%, +2.8\%]$ in the baseline regression and -0.5% a year $[-3.6\%, +2.9\%]$ in the extended regression, so the 90% intervals rule out unexplained trends larger than about 4% a year in either direction (Figure 3b). Pure affordability predicts an unexplained trend of zero, so an interval that includes zero is the predicted outcome. The spending slope β_t shows no trend from 1994 to 2012 either, as prediction 2 requires (Appendix A.1).

Because the price effect is about one half, pure affordability predicts a much faster spread of

Figure 3. A 10% fall in the quality-adjusted computer price lowered the spending threshold by 4–6%, about half the fall that pure affordability predicts, and the spending threshold showed no other steady change



The price effect b and the unexplained trend d on a percentage scale, from the 20 yearly spending thresholds of 1994–2013, main computer price: posterior medians with 90% intervals. (a) The fall in the spending threshold when the quality-adjusted computer price falls 10%, $100\,(1 - 0.9^b)$ percent. Pure affordability ($b = 1$) predicts a fall of 10% (dotted line). (b) The unexplained trend, the yearly change in the spending threshold that the computer price does not explain, $100\,(e^d - 1)$ percent a year. Pure affordability predicts 0% (dotted line). Each panel shows the baseline regression and the extended regression, and Figure A2 in Appendix A shows every other specification, including the two intermediate fits and the joint model.

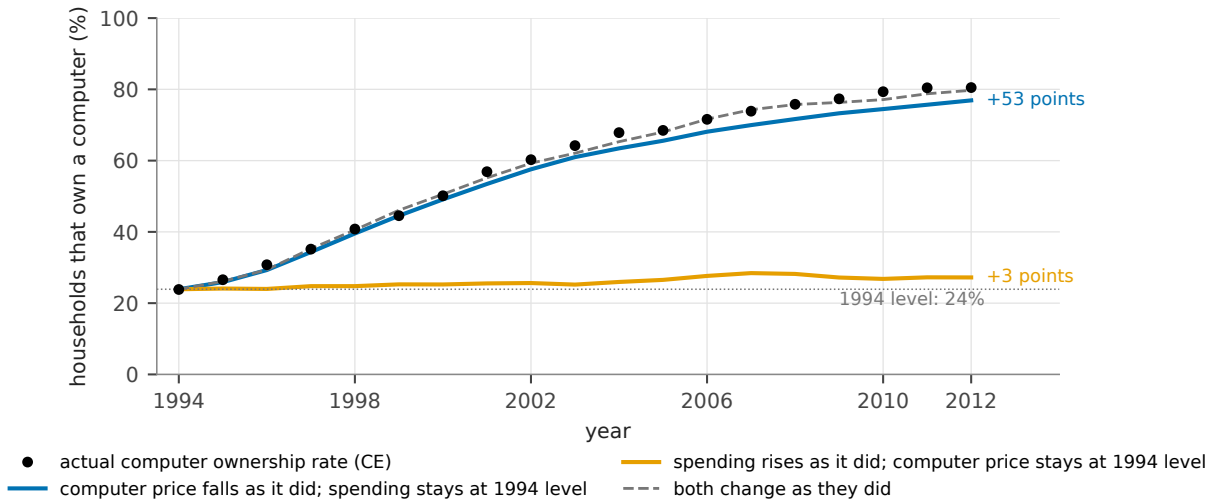
computers from 1994 to 2012 than occurred. Applied to 2012 households at the 2012 computer price, the 1994 probit model predicts that *nearly every household* owned a computer by 2012, against the actual 81% (Appendix B.2).

Even at half strength, falling prices raised computer ownership far more than rising incomes did. Falling computer prices alone would have raised the computer ownership rate from 24% in 1994 to 77% in 2012, close to the actual 81%, while rising household spending alone would have raised it only to 27%. Figure 4 shows what the computer ownership rate would have been in each year from 1994 to 2012 if only one of the two forces had changed after 1994, according to the baseline regression. Both forces are measured in constant 1994 dollars. In the blue path, the quality-adjusted computer price falls as it did, and household spending keeps its 1994 distribution. In the orange path, household spending rises as it did after inflation, and the computer price stays at its 1994 level relative to all consumer prices. With both forces at work, the model's path (dashed) stays within 2.6 percentage points of the actual computer ownership rate in every year from 1994 to 2012.

The computer price did most of the work because it changed far more than household spending did. From 1994 to 2012 the quality-adjusted computer price, relative to all consumer prices, fell to 1/79 of its 1994 level. Over the same years the long-run spending of the median household rose by 14% after inflation, from \$19,200 to \$21,900 a year in 1994 dollars. Because the price effect is about one half, a 1% rise in household spending did about as much for computer ownership as a 2% fall in the computer price, but the computer price fell *much further* than household spending rose.

The fall in the computer price accounts for 52.8 percentage points of the 1994–2012 rise in the computer ownership rate, or 94% of the part that prices and incomes explain (93% to 96% across the 90% interval for b), and the growth of household spending accounts for 3.1 points, or 6%. These shares average over the two orders in which the two

Figure 4. Falling computer prices alone would have raised the computer ownership rate from 24% in 1994 to 77% in 2012, and rising household spending alone only to 27%



The computer ownership rate of US households, 1994–2012, if only the computer price or only household spending had changed after 1994 (baseline regression, price effect $b = 0.45$, constant 1994 dollars). Blue: the quality-adjusted computer price, relative to all consumer prices, falls as it did, and household spending keeps its 1994 distribution. Orange: household spending rises as it did after inflation, and the computer price stays at its 1994 level relative to all consumer prices. Dashed: both change as they did. Dots: the actual computer ownership rate of CE households, with survey weights. Every path starts from the 1994 spending threshold and spending slope (Section 4.3).

changes can be made (a Shapley split, Section 4.3). A split in current dollars counts general inflation as growth in household spending, and that split gives household spending 19% of the part of the 1994–2012 rise that prices and incomes explain (Online Appendix E.1).

Quality improvement, more than lower dollar prices, paced the spread of computers. Most of the fall in the quality-adjusted computer price from 1994 to 2012 came from better computers, and little of it from lower dollar prices. Figure A3 in Appendix A.4 plots each computer price, monthly, relative to its 1994 average. From 1994 to 2012 the quality-adjusted computer price fell to 1/56 of its 1994 level (by 4.02 in logs). From 1994 to 2012 the dollar price that CE households paid for a computer fell by only 7% at the median and by 47% at the 90th percentile. The dollar price of a computer equals its quality-adjusted price times its quality, so the fall in the quality-adjusted price splits into a rise in quality and a fall in the dollar price. Measured in logs, better computers account for 84–98% of the 1994–2012 fall in the quality-adjusted computer price, and lower dollar prices account for the rest. The dollar price fell far less than the quality-adjusted price because the gains in technology went into better computers and into productization, the work of turning the computer into a product that ordinary households could buy and use. The median price paid is a noisy series that rose after 2010, which may partly reflect the CE's move of computer accessories to a code of their own in the second quarter of 2011. The 90th percentile is steadier, so the lower bound of 84% is the firmer number.

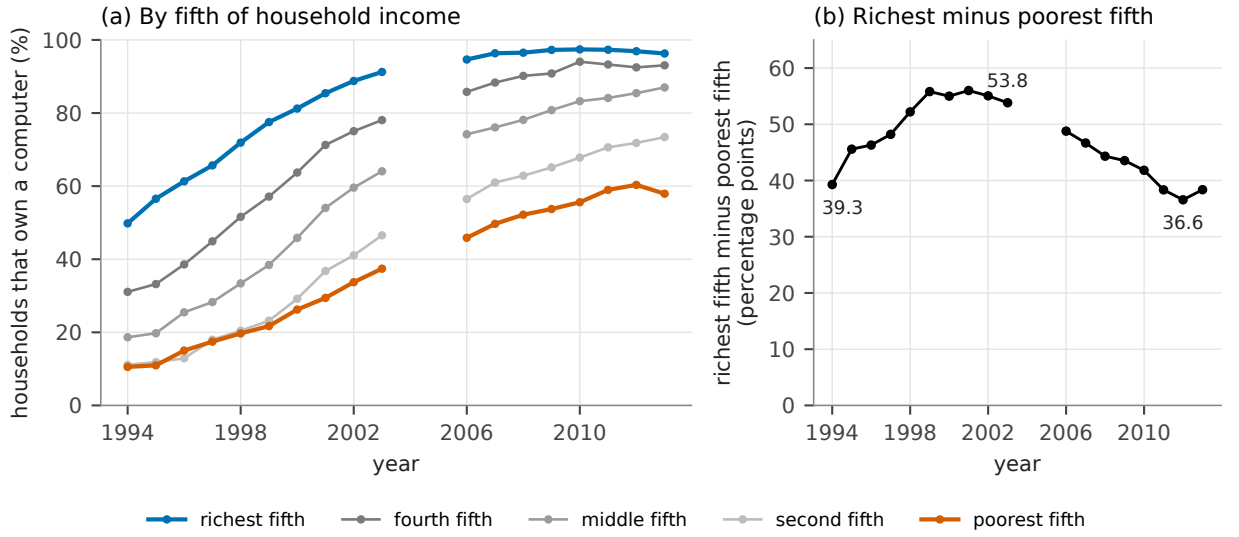
The other quality-adjusted computer price indexes give similar price effects, but the dollar price that households paid explains almost none of the fall in the spending threshold. The two other BEA price indexes give price effects between 0.39 and 0.79 over 1994–

2013, and the CPI-U for personal computers, which covers only 1999–2013, gives 0.63 to 0.93 (Appendix A.4). Measured against the median price paid, the computer price explains almost none of the fall in the spending threshold from 1994 to 2012, 81% in the baseline regression and 89% in the extended regression, and an unexplained trend of 8–12% a year accounts for nearly all of that fall (Online Appendix E.2).

5.2 As computer prices fell, the gap in computer ownership between rich and poor households widened, then narrowed

The gap between the shares of the richest and poorest fifths of households that owned a computer widened from 39.3 percentage points in 1994 to 53.8 in 2003, and then narrowed to 36.6 in 2012, slightly below its 1994 level (Figure 5b). Figure 5a shows the computer ownership rate in each fifth of reported household income, in each year from 1994 to 2013 except 2004 and 2005, whose CE files carry no income ranking. From 1994 to 2003 the computer ownership rate of the richest fifth rose by 41 percentage points, from 49.8% to 91.2%, and the computer ownership rate of the poorest fifth rose by 27 points, from 10.5% to 37.4%. From 2003 to 2012 the richest fifth gained 6 points, to 96.9%, and the poorest fifth gained 23 points, to 60.3%. The gains in computer ownership moved down the income distribution, as the spending threshold did. *Poorer households gained from falling computer prices last.* The poorest fifth of households matched the richest fifth's 1996 computer ownership rate (61.3%) only in 2012 (60.3%), 16 years later.

Figure 5. The gap in computer ownership between the richest and poorest fifths of households widened from 1994 to 2003 and then narrowed, and the poorest fifth reached the richest fifth's 1996 computer ownership rate only in 2012



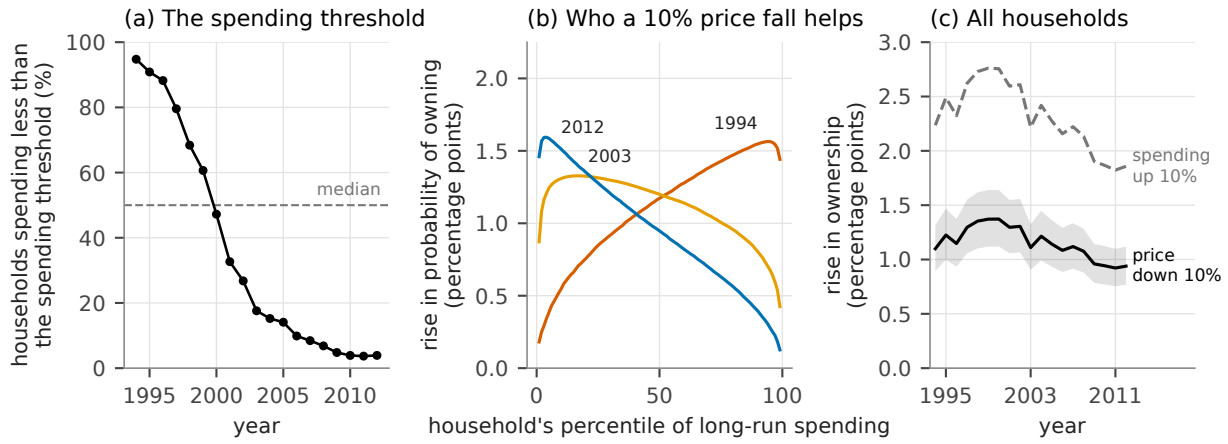
Computer ownership by fifth of household income, CE, 1994–2013, with survey weights. (a) The share of households that own a computer in each fifth of reported income, among CE interviews with the appliance inventory and a complete income report, with households ranked by the CE's own income ranking. The CE files for 2004 and 2005 carry no income ranking. (b) The computer ownership rate of the richest fifth minus the computer ownership rate of the poorest fifth, in percentage points.

This gap in computer ownership between rich and poor households widened from 1994 to 2003 and narrowed from 2003 to 2012 because, over those years, the spending threshold fell *from the top of the distribution of household spending to the bottom.* Among households with the same location and traits, the falling spending threshold passed the spending of richer households first and the spending of poorer households last. Figure 6a places each year's spending threshold in that year's distribution of long-run spending. In 1994, 95% of households spent less than the spending threshold, so only the top 5% of households were more likely than not to own a computer. The spending threshold passed the spending of the median household between 1999 and 2000, as the computer ownership rate passed 50%. By 2012 only 4% of households spent less than the spending threshold.

Every household faced the same computer price, but from 1994 to 2012 the households that a price cut helped most moved from the top of the income distribution to the bottom. Figure 6b shows how much a 10% fall in the quality-adjusted computer price raised a household's probability of owning a computer, by the household's percentile of long-run spending. In 1994 the rise was largest near the top of the spending distribution, 1.55 percentage points at the 90th percentile against 0.55 at the 10th percentile. In 2012 the rise was largest near the bottom, 1.51 points at the 10th percentile against 0.39 at the 90th percentile. In the 1990s, when price cuts mostly helped high-income households, the gap in computer ownership between rich and poor households widened. After 2003, when most high-income households already owned a computer and price cuts mostly helped low-income households, the gap in computer ownership narrowed.

A 10% cut in the computer price brought in the most new computer owners in 1999 and 2000, when the spending threshold stood near the spending of the median household and the most households had spending near the spending threshold. Figure 6c averages over households. The 10% cut raised the computer ownership rate by 1.10 percentage points $[0.89, 1.32]$ in 1994, by 1.37 points in 1999 and 2000, and by 0.94 points $[0.77, 1.12]$ in 2012. From 2000 on the spending threshold lay in the bottom half of the spending distribution, so the number of new computer owners that each price cut brought in depended on the incomes of the bottom half of households.

Figure 6. From 1994 to 2012 the households that a cut in the computer price helped most moved from the top of the spending distribution to the bottom



Household spending, the computer price, and their interaction (baseline regression, main computer price, $b = 0.45$). (a) Share of households whose long-run spending is below the spending threshold $\hat{\tau}_t$, in each year from 1994 to 2012, the log long-run spending at which a household with average location and traits has a 50% chance of owning a computer. (b) Rise in a household's probability of owning a computer from a 10% fall in the quality-adjusted computer price, $\Phi\big(\hat{\beta}_t(x - \hat{\tau}_t + b\ln(1/0.9))\big) - \Phi\big(\hat{\beta}_t(x - \hat{\tau}_t)\big)$, by the household's percentile of long-run spending, for a household in the pooled average geography, in 1994, 2003, and 2012. (c) The same rise averaged over households in each year from 1994 to 2012, with a 90% band from b alone, and the rise from a 10% rise in every household's spending.

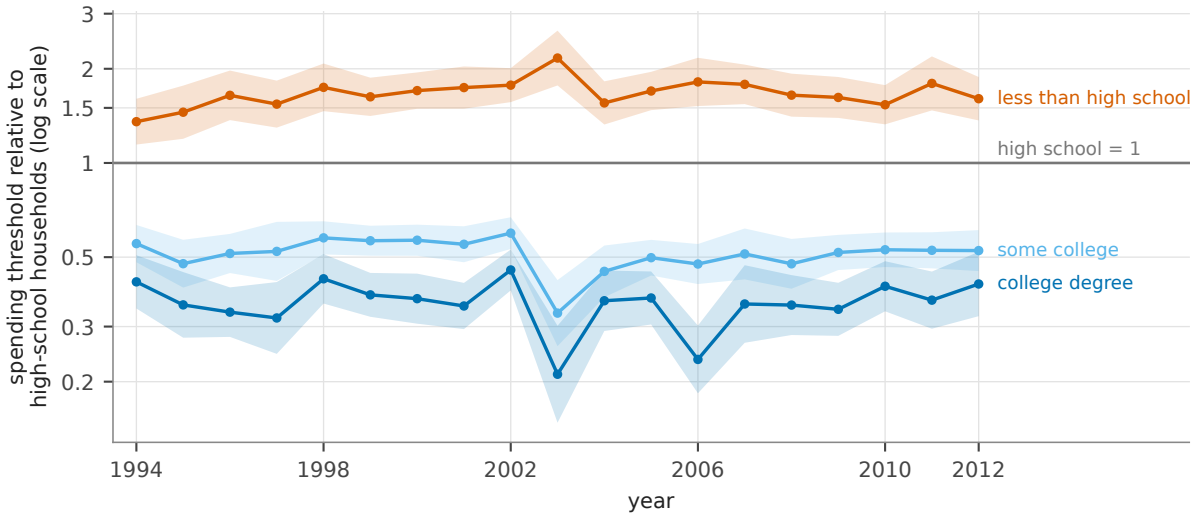
5.3 Falling computer prices still left the education gap in computer ownership in place

In the CE data, with survey weights and no model, the gap in computer ownership between households headed by college graduates and households headed by high-school graduates widened as computer prices fell in the 1990s and narrowed only after 2001. The gap between the two groups' computer ownership rates was 29.8 percentage points in 1994 and 39.3 in 1997, stayed above 36 points through 2001, and then fell to 22.9 in 2012. As with the gap in computer ownership between rich and poor households, the education gap in computer ownership narrowed *only once most households in the leading group already owned a computer*, which for college households was from 2001 on, when more than 80% of them did. The gap in computer ownership between college households and households whose head did not finish high school grew from 38.0 percentage points in 1994 to 59.4 in 2001 and was still 43.8 in 2012.

Among households with the same long-run spending, the education gap in computer ownership did not shrink as the computer price fell. Figure 7 shows, for each year from 1994 to 2012, the spending threshold of households in each education group of the extended regression, relative to the spending threshold of otherwise similar households headed by a high-school graduate. Averaged over 1994–2012, the spending threshold of households headed by a college graduate was about one third (0.35) of the spending threshold of households headed by a high-school graduate. **A household headed by a college graduate that spent $10,000 a year was as likely to own a computer as an otherwise similar household headed by a high-school graduate that spent $28,000, or 2.8 times as much.** The multiple was 2.4 in 1994 and again in 2012,

and a weighted linear trend puts its change at +0.2% a year [−0.9%, +1.3%]. The college ratio lies between 0.32 and 0.45 in every year except 2003 and 2006 (0.21 and 0.24), the two years with the flattest spending slope. Averaged over 1994–2012, households headed by someone with some college had about half the spending threshold of high-school households (0.51), and households headed by someone who did not finish high school had 1.67 times the spending threshold of high-school households, and these differences did not narrow either. *Prediction 3 fails.* At the same long-run spending, more educated households owned computers earlier.

Figure 7. Among households with the same spending, the education gap in computer ownership did not shrink as computer prices fell from 1994 to 2012



Spending thresholds by education of the household head, extended regression, 1994–2012, relative to the spending threshold of otherwise similar households headed by a high-school graduate, with 90% intervals (±1.645 replicate standard errors), on a log scale. Each line is $\exp(-\theta_{e,t})$ for education level e in year t. A value of 0.35 means that a household with that education was as likely to own a computer as a household headed by a high-school graduate that spent $1/0.35 = 2.8$ times as much. The education shift is $\theta_{e,t} = \delta_{e,t}/\beta_t$ (Section 4.1), so in 2003 and 2006, the two years with the flattest spending slope β_t, every education shift is larger.

The gap in computer ownership between college and high-school households could partly reflect income that long-run spending misses. A control function, which uses current income as an instrument for long-run spending (Appendix A.2), shrinks the college shift, averaged over 1994–2012, by about a quarter. With the control function, a household headed by a college graduate was still as likely to own a computer as a household headed by a high-school graduate that spent 2.2 times as much (Appendix A.3).

5.4 Robustness

The estimates hold up in the robustness checks of Appendix B. Fitting the extended regression's two stages at once, as one Bayesian model of all 178,891 CE households of 1994–2013, gives $b = 0.60$ $[0.45, 0.74]$ and $d = -0.003$ $[-0.038, 0.029]$, against 0.59 and −0.005 from the two stages. Under most other choices of sample, spending measure, and link function, b stays between 0.37 and 0.61, and in constant dollars b is within 0.02 of its current-dollar value. Two checks give larger price

effects. Counting vehicle purchases as spending gives 0.88 [0.71, 1.05] in the extended regression, and replacing long-run spending with current income, a measure of long-run resources with much noise, gives 1.38 [0.86, 1.95]. Apart from current income, d stays between -0.014 and $+0.031$ a year. The second-stage model fits the yearly spending thresholds, and the two-stage method recovers known values of b and d from fake data.

6 Discussion

What the results say about the profit switch. The results support the affordability phase of the profit switch hypothesis in part. From 1994, when 24% of US households owned a computer, to 2012, the spending threshold moved in step with the quality-adjusted computer price and showed no other steady change, so affordability matches the timing of the spread of computers over those years. But the spending threshold fell only about half as fast as the computer price, in percentage terms, and education shifted the spending threshold at a given level of spending, so affordability alone does not account for which households owned a computer. The first phase of the hypothesis, in which firms' profits and investment pace the spread of a technology, lies before spring 1992, when the computer ownership rate passed 20%, and only two surveys, the CPS supplements of October 1984 and October 1989, measure computer ownership in those years. The surveys therefore cannot test whether the forces behind computer ownership switched between the two phases. When we fit the CPS and CE spending thresholds together, the unexplained trend appears to change in late 1995, when the computer ownership rate passed 30%, but that change falls where the data move from the CPS to the CE, and within the CPS alone a model with one unexplained trend predicts best (Appendix B.3).

Why the price effect is about one half. The household model offers two readings of a price effect of about 0.5. In the first reading, other costs of owning a computer, such as software, an internet connection, and the time it takes to learn to use a computer, fell more slowly than the hardware price and made up about half of the cost of ownership. In the second reading, households valued only part of the quality improvement that the price index counts. A faster processor counts fully in the price index but may matter little to a household that uses its computer for email and letters. *Both readings say that the effective cost of owning a computer fell about half as fast, in percentage terms, as the quality-adjusted computer price.* Our data cannot tell the two readings apart. The size of the price effect also depends on which quality-adjusted index is used, and the BEA indexes that fell less than the main index give price effects of up to 0.79 over 1994–2013 (Appendix A.4). Measures based on computing performance show larger price falls than the official indexes (Nordhaus, 2007), and against such measures the price effect would be smaller still.

What the absence of an unexplained trend means. An unexplained trend near zero means that, nationally, *nothing other than the computer price* changed the spending threshold steadily from 1994 to 2013. Spillovers from learning or from networks that grew steadily with computer ownership would have made the spending threshold fall faster than the computer price explains, which would show up as a negative unexplained trend. The estimates rule out unexplained trends larger than about 4% a year, and the fake-data check of Appendix A.2 shows that an unexplained trend of about 5% a year would have been detected. Smaller unexplained trends remain possible, and because the estimates of b and d are tied, a negative unexplained trend would come with an

even smaller price effect. Local spillovers of the kind that Goolsbee and Klenow (2002) find need not show up as a national trend.

What the education gap means. Among households with the same long-run spending, the education gap in computer ownership was large from 1994 to 2012 and *did not shrink* as computer prices fell, even though the same gap in the raw survey data narrowed after 2001 as most college households came to own a computer. Part of the education gap in computer ownership may reflect income that long-run spending misses, since the control function shrinks this gap by about a quarter. The rest of the education gap suggests that, at the same resources, more educated households valued a computer more, found a computer easier to use, or knew more about computers. Our data cannot separate those readings.

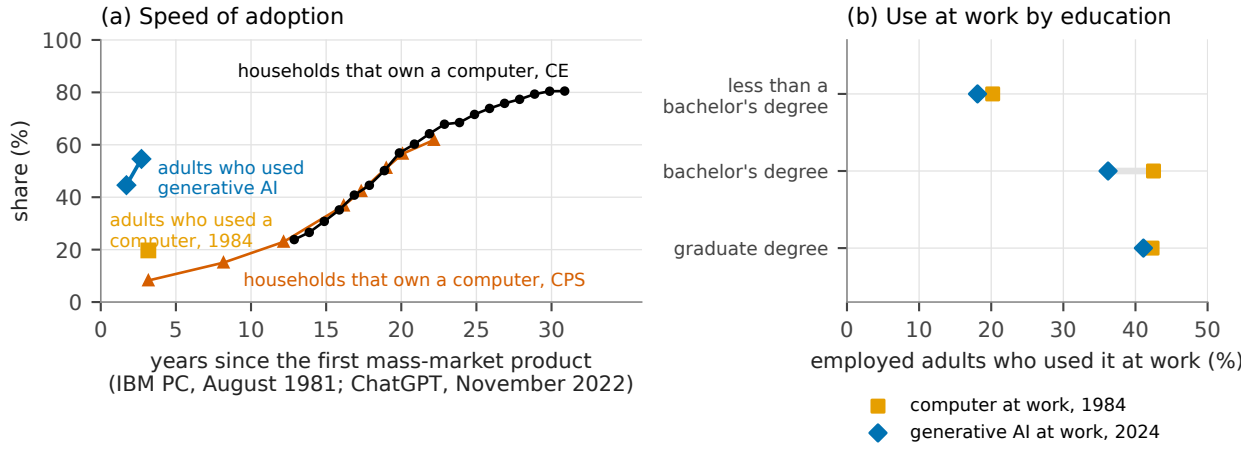
How the findings compare with earlier work. Our findings fit the closest earlier work, and they add what falling computer prices changed and what they left in place. Chinn and Fairlie (2007) find that differences in income per head account for most of the gap in computer and internet use across 161 countries in 1999–2001. We find that falling prices account for 94% of the 1994–2012 rise in US computer ownership that prices and incomes explain. The two findings need not conflict. Chinn and Fairlie (2007) compare countries at one point in time, when incomes differ widely across countries, while we follow one country over the 18 years from 1994 to 2012, in which the quality-adjusted computer price fell to 1/56 of its 1994 level and the long-run spending of the median household rose by 14% after inflation. The order in which income groups bought computers matches the finding of Greenwood et al. (2005, p. 111) that poor households bought durable goods later than rich households, when prices were lower. Our household model adds why the gap in computer ownership between rich and poor households widened from 1994 to 2003. A price cut does the most for households close to buying, and in 1994 the households close to buying a computer were the richest households. The education gap in computer ownership matches evidence on computer use at work (Riddell and Song, 2017) and on generative AI. Bick et al. (2024) find that generative AI and the personal computer had similar early adoption patterns by education (Section 7). We add that, at the same long-run spending, the education gap in home computer ownership did not shrink from 1994 to 2012, while the quality-adjusted computer price fell 98%.

Limits. Computer prices are the same for every household in a given year, so the price effect and the unexplained trend rest on the 20 yearly spending thresholds of 1994–2013. The estimates of the price effect and the unexplained trend therefore also pick up anything else that moved with computer prices over time, including the spread of the internet and of software, and network effects among computer users (Goolsbee and Klenow, 2002). The household data can measure an unexplained trend, but the household data cannot say what caused an unexplained trend. The early phase of the spread of computers is thinly observed, since only the CPS supplements of October 1984 and October 1989 measure computer ownership while the computer ownership rate was below 20%, and those two supplements record current income in brackets. The CE computer ownership question changed its wording in the second quarter of 2001, and the CPS education question changed between 1989 and 1993, mainly between high school and some college (Jaeger, 1997). We estimate the mapping from income to spending in 1994–1998 and apply the mapping to every CPS year, and the 2013 spending thresholds rest on one quarter of data.

7 Implications for AI diffusion and policy

Generative AI has spread faster than the personal computer did. Figure 8a counts years from the first mass-market product of each technology, the IBM PC in August 1981 and ChatGPT in November 2022, the dates that Bick et al. (2024) use. In August 2025, less than three years after the release of ChatGPT, 54.6% of US adults aged 18 to 64 used generative AI, up from 44.6% in August 2024 (Bick et al., 2025). In 1984, three years after the IBM PC went on sale, 19.7% of US adults aged 18 to 64 used a personal computer (Bick et al., 2025), and 8.2% of US households owned a computer. The share of US households that owned a computer passed 50% only in 2000, nineteen years after the IBM PC went on sale.

Figure 8. Generative AI has spread faster than the personal computer did, and for both technologies workers with a bachelor's degree were about twice as likely as other workers to use the technology at work



Home computers and generative AI in the United States. (a) Adoption by years since each technology's first mass-market product, the IBM PC (August 1981) and ChatGPT (November 2022). Black and red: the share of households that own a computer, from the CE (yearly averages, 1994–2012) and the CPS supplements (October 1984 to October 2003). Orange: the share of adults aged 18 to 64 who used a personal computer in 1984. Blue: the share of adults aged 18 to 64 who used generative AI in August 2024 and August 2025, from the Real-Time Population Survey (RPS) of Bick et al. (2024). The orange and blue points are published estimates (Bick et al., 2025). (b) The share of employed adults aged 18 to 64 who used a computer at work in October 1984 (CPS) and who used generative AI at work in August and November 2024 (RPS), by education (Bick et al., 2024, Figure 5a).

As with the computer, more educated adults adopted generative AI first (Figure 8b). In October 1984, 42.5% of employed adults aged 18 to 64 with a bachelor's degree used a computer at work, against 20.2% of employed adults without a bachelor's degree. In August and November 2024, 36.2% of employed adults with a bachelor's degree used generative AI at work, against 18.1% of employed adults without a bachelor's degree (Bick et al., 2024, Figure 5a). For both technologies, workers with a bachelor's degree were about twice as likely as workers without one to use the technology at work.

Much of the use of generative AI costs households *nothing in dollars*. Bick et al. (2024) point to low adoption costs as one likely reason that generative AI spread faster than the personal computer outside work, since a computer required expensive hardware while generative AI is available free

or with a low-cost subscription. The basic tiers of the main AI assistants are free, and the paid tiers are subscriptions at fixed dollar prices. Anthropic's Claude Pro subscription, for example, cost $20 a month when it launched in September 2023 (Anthropic, 2023), and Claude Pro still cost $20 a month in November 2026 (Anthropic, 2026). The adoption rates in Figure 8 count any use of generative AI, paid or free.

Firms can also gain from AI use indirectly, through advertising and through users' data. OpenAI said in January 2026 that it would test advertising in the United States in the free tier of ChatGPT and in its low-cost Go plan, while keeping its higher-priced plans free of ads (OpenAI, 2026), and since 2025 Anthropic has asked users of its free and paid consumer plans whether their conversations may be used to train its models (Anthropic, 2025). In the terms of the profit switch hypothesis, free tiers are one way that firms' profits and investment pace the early adoption of AI, and advertising and data may let firms keep AI free to many households long after the early phase. How far AI funded in these ways spreads may then depend more on what firms earn from it than on what households can spend. The household model of this paper compares what a household can spend with a dollar price, so the implications below hold *only for AI that households pay for in dollars*, such as subscriptions.

If paid AI services follow the pattern that home computers followed after 1994, when 24% of US households owned a computer, **the spread of paid AI may follow affordability measured against a quality-adjusted price of AI services**. Paid consumer AI has mostly been sold at fixed monthly prices while the capability of AI rose. From 1994 to 2012 the dollar price that households paid for computers fell far less than the quality-adjusted computer price, which fell to 1/56 of its 1994 level, and measured against the dollar price, affordability explains almost none of the spread of computers. The subscription price of AI may say as little about the spread of paid AI. If the price effect estimated for computers over 1994–2013 carries over to AI, **a 10% fall in the quality-adjusted price of AI services would lower the spending threshold for paid AI by 4–6%**, and a 1% rise in household spending would do as much for paid AI adoption as a fall of about 2% in that price. For computers, the quality-adjusted price fell so far from 1994 to 2012 that falling prices account for 94% of the rise in computer ownership over those years that prices and incomes together explain, so if the quality-adjusted price of AI services falls as fast, income growth would set little of the pace of paid AI adoption.

Every household would face the same price for a paid AI service, but **a fall in that price would be most likely to tip households close to paying for AI into paying for it**, the households whose spending is near the spending threshold for paid AI. At first these would be high-income households, and as the spending threshold falls they would be lower-income households, so the gap between the shares of rich and poor households that pay for AI may first widen and then narrow, as the gap in computer ownership did. For computers, the poorest fifth of households matched the richest fifth's 1996 computer ownership rate, about 60%, only in 2012, 16 years later. **Once the spending threshold for paid AI falls below the spending of the median household, how many households each price cut brings to paid AI would depend on the incomes of the bottom half of households.** This prediction offers a household mechanism for the companion cross-country finding that, above a switch point of about 5% of the population, internet adoption rose faster in countries with a larger bottom-50% income share (an association across 22 advanced economies, 1991–2016). Cheaper AI may do less to narrow the gap in paid AI use between more and less educated households. For computers, averaged over 1994–2012, a household headed by a college graduate was as likely to own a computer as an otherwise similar household headed by a high-school graduate that spent 2.8 times as much, and this multiple did not shrink as computer

prices fell, so **falling prices may leave gaps in paid AI adoption between more and less educated households with the same income in place**.

For policy, the first need is measurement. **Policymakers will need a quality-adjusted price index for AI services, of the kind that the BEA and BLS publish for computers, to track how affordable AI is becoming.** Testing the profit switch hypothesis or its affordability phase for AI also requires a household survey that records AI use, paid or free, together with income or spending from early adoption on, ideally every year. The CPS computer supplements began in 1984, when 8% of households owned a computer, but ran only every four or five years before 1997. With both kinds of data, the two stages of this paper can be run below and above 20–30% adoption, and the price effect and the unexplained trend can be compared across the switch.

Unless policy subsidizes lower-income households' access to paid AI and helps them learn to use it, paid AI may reach those households more than a decade after richer households, as home computers did. Price cuts alone may leave lower-income households the last to pay for AI, because a price cut does most for the households closest to affording paid AI, and at first those households would be richer households. A subsidy aimed at lower-income households would let them pay for AI before price cuts bring paid AI within their reach. A price subsidy alone may not be enough. In one reading of the price effect of about one half (Section 6), half of the cost of owning a computer lay outside its price, in software, an internet connection, and the time it takes to learn to use a computer, and cheaper AI alone may leave the education gap in paid AI use in place (Section 5.3).

Once the spending threshold for paid AI is below the spending of the median household, a given amount of income support or price subsidy would raise paid AI adoption more if it went to the bottom half of households than if it were spread across all households, because the households close to adopting would then be lower-income households.

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Appendices A and B are part of the paper. Online Appendices C to I describe how we built the dataset and give the method and the remaining estimates in detail. The online appendix is at https://github.com/IlanStrauss/home-computer-diffusion/blob/main/writeup/online_appendix.pdf.

A Robustness

The price effect b stays well below 1, the value that pure affordability predicts, and the unexplained trend d stays near zero, from the baseline regression to the extended regression and in most of the other checks.

A.1 From the baseline regression to the extended regression

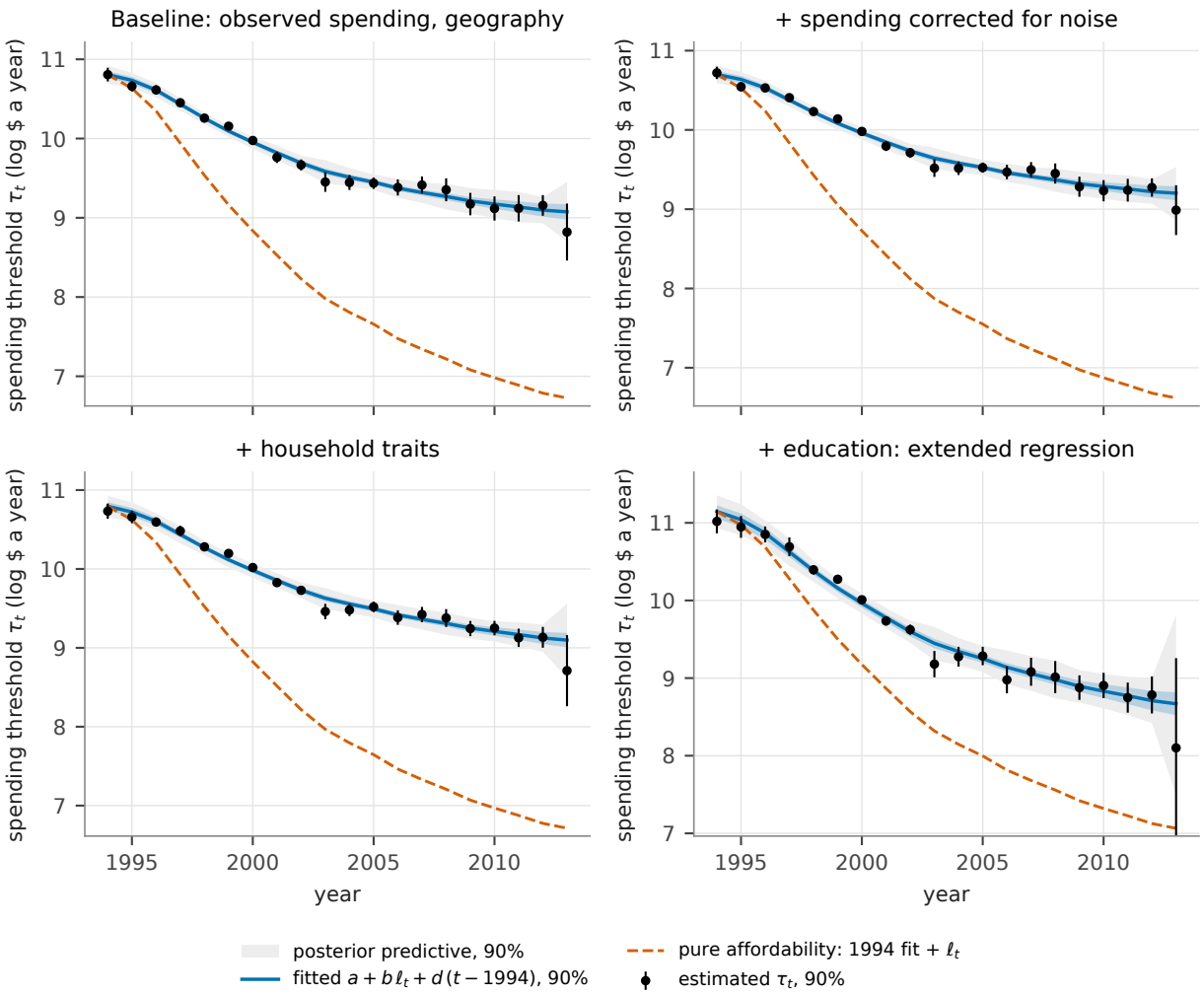
As the extended regression adds its three parts to the baseline regression, the price effect b moves between 0.39 and 0.59, and every 90% interval for b lies below 1. Table A1 gives the second-stage estimates under the main computer price, and Figure A2 plots b and d in every specification. In the baseline regression, $b = 0.45$ $[0.37, 0.54]$ and $d = +0.005$ $[-0.016, 0.028]$. Correcting spending for noise lowers b to 0.39, adding household traits barely changes b (0.41), and adding education, which completes the extended regression, raises b to 0.59 $[0.46, 0.72]$. Every interval for b excludes 1. In all four CE regressions, d lies between -0.005 and $+0.005$ a year. Over the 18 years from 1994 to 2012, the unexplained trend adds between -0.08 and $+0.10$ to the log spending threshold, against a fall of 1.56 to 2.36 from the computer price. So, in current dollars and against the main computer price, the path of the spending threshold is a scaled-down copy of the path of the computer price. The 20 yearly spending thresholds of 1994–2013 cannot rule out unexplained trends smaller than about 4% a year, because the estimates of b and d trade off against each other (Section 4).

The estimates of b move across the four CE regressions with the spending slope β_t. The spending threshold is measured along the spending axis, so when the probability of owning a computer rises less steeply with spending, a given year-to-year rise in the probability of owning a computer at fixed spending moves the spending threshold further along the spending axis. The spending correction steepens the spending slope (0.79 to 1.04, against 0.70 to 0.87 in the baseline regression) and lowers b. Education takes over part of the link between spending and computer ownership, flattens the spending slope (0.52 to 0.78), and raises b.

Across the four CE regressions the log spending threshold fell by 1.45 to 2.24 from 1994 to 2012, while the log main computer price fell by 4.02. In the baseline regression the log spending threshold is 10.807 in 1994 ($49,400 a year), 9.452 in 2003 ($12,700), and 9.156 in 2012 ($9,500), and Online Appendix Table E1 gives every year from 1994 to 2013. The 2013 spending thresholds rest on one quarter of data and have standard errors of 0.19 (corrected spending alone) to 0.70 (extended regression), so the tables report changes to 2012.

The spending slope shows no trend, as prediction 2 requires. In the baseline regression, β_t is 0.70 to 0.87 in every year from 1994 to 2012, and in the extended regression, β_t is 0.52 to 0.78, with its low points in 2003 and 2006. From 1994 to 2012 the log spending threshold fell 1.8 to 2.6 less than the log computer price, so by 2012 the spending threshold was 6 to 13 times the level that pure affordability predicts (Appendix B.2).

Figure A1. In each of the four CE regressions, the spending threshold fell with the computer price but far less than pure affordability predicts



Yearly spending thresholds $\hat{\tau}_t$, 1994–2013, in the baseline regression and as the extended regression adds its three parts one at a time, with 90% intervals (±1.645 replicate standard errors). Lines and bands as in Figure 2: the fitted second stage (blue), its 90% posterior predictive interval (grey), and the path that pure affordability predicts from the fitted 1994 spending threshold (dashed). Main computer price (BEA PCE price index for personal computers). A log spending of 10 is $22,000 a year and a log spending of 9 is $8,100.

Table A1. Second stage under the main computer price: posterior medians with 90% intervals. a is the 1994 log spending threshold, b the price effect (the change in the log spending threshold per unit change in the log computer price), d the unexplained trend (the yearly change in the log spending threshold that the computer price does not explain), and ω the standard deviation of year-specific deviations from the fitted line. The last three columns split the observed change in the log spending threshold from 1994 to 2012 into the part due to the computer price ($\ell_{2012} - \ell_{1994} = -4.02$) and the part due to the unexplained trend. The lower panel allows the sampling errors of the yearly spending thresholds to be correlated across years (Appendix A.2).

Specification	a	b	d	ω	$\hat{\tau}_{2012} - \hat{\tau}_{1994}$	$b\,(\ell_{2012} - \ell_{1994})$	$18\,d$
Baseline	10.81	0.45	+0.005	0.029	−1.65	−1.81	+0.10
	[10.76, 10.85]	[0.37, 0.54]	[−0.016, +0.028]	[0.004, 0.060]		[−2.16, −1.47]	[−0.28, +0.50]
+ corrected spending	10.70	0.39	+0.005	0.034	−1.45	−1.56	+0.08
	[10.65, 10.74]	[0.31, 0.47]	[−0.015, +0.026]	[0.009, 0.065]		[−1.90, −1.25]	[−0.28, +0.46]
+ household traits	10.79	0.41	−0.002	0.043	−1.60	−1.63	−0.04
	[10.74, 10.85]	[0.32, 0.49]	[−0.023, +0.019]	[0.021, 0.073]		[−1.97, −1.29]	[−0.41, +0.34]
+ education (extended)	11.15	0.59	−0.005	0.062	−2.24	−2.36	−0.08
	[11.05, 11.23]	[0.46, 0.72]	[−0.037, +0.029]	[0.028, 0.108]		[−2.89, −1.83]	[−0.66, +0.52]
Sampling errors correlated across years (covariance check)							
Baseline	10.81	0.45	+0.007	0.028	−1.65	−1.80	+0.12
	[10.75, 10.86]	[0.35, 0.55]	[−0.018, +0.032]	[0.007, 0.058]		[−2.19, −1.42]	[−0.33, +0.58]
+ household traits	10.80	0.41	0.000	0.043	−1.60	−1.65	0.00
	[10.73, 10.86]	[0.32, 0.51]	[−0.023, +0.024]	[0.022, 0.074]		[−2.03, −1.28]	[−0.41, +0.42]
+ education (extended)	11.15	0.59	−0.003	0.059	−2.24	−2.37	−0.05
	[11.05, 11.24]	[0.45, 0.73]	[−0.038, +0.032]	[0.027, 0.105]		[−2.94, −1.80]	[−0.68, +0.58]

A.2 Other checks

We checked the estimates as the Bayesian workflow of Gelman et al. (2020) recommends.

The second-stage model fits the yearly spending thresholds. In a posterior predictive check, which asks whether data simulated from the fitted model look like the observed data, every observed spending threshold lies inside its 90% predictive interval in the baseline regression and with corrected spending. Once household traits enter, and in the extended regression, every spending threshold except the 2003 spending threshold lies inside its 90% predictive interval. The year-specific deviations from the fitted line are small ($\omega = 0.03$ to 0.06, or 3–6%).

Allowing the sampling errors of the yearly spending thresholds to be correlated across years barely changes b and d. The CE reuses its sampling areas across years, so the replicate estimates of the yearly spending thresholds are positively correlated, and the correlation falls with the gap between years (baseline regression: 0.40 for adjacent years, 0.15 for years five apart). We fit a correlation $R_{st} = r\,\rho^{|s-t|}$ between the sampling errors of years s and t ($r = 0.47$ and $\rho = 0.82$ in the baseline regression, and 0.18 and 0.82 in the extended regression). When this correlation enters the likelihood, b and d move by at most 0.004 (Table A1, lower panel).

Measuring in constant dollars changes the unexplained trend a little and the price effect hardly at all. With the spending thresholds and the computer price divided by the PCE price index for all items, b is 0.37 to 0.58, within 0.02 of the current-dollar value of b, and d is between −0.017 and −0.009 a year, with every 90% interval including zero (`second_stage_constant.R`). The unexplained trend is lower in constant dollars because, in current dollars, the spending threshold moved with the computer price alone, and general inflation did not raise the spending threshold. In constant dollars the spending threshold therefore fell by about $(1 - b)$ times the inflation rate more than the computer's relative price explains.

Other choices of sample, spending measure, and link function give similar answers (Table A2 and

Figure A2). Under every check, b stays between 0.37 and 0.61, with two exceptions. The price effect is higher when vehicle purchases count as spending in the extended regression (0.88 $[0.71, 1.05]$), and much higher when current income replaces long-run spending (1.38 $[0.86, 1.95]$). The current-income estimate is not comparable with the other estimates. Current income measures long-run resources with much noise. The noise in current income flattens the spending slope (a mean $\hat{\beta}_t$ of 0.21, against 0.74 with corrected spending) and stretches the path of the estimated spending threshold, as Online Appendix C.4 predicts. Under current income the unexplained trend is -0.171 $[-0.297, -0.037]$, against -0.013 with corrected spending for the same households. A control function (Rivers and Vuong, 1988) uses current income in a different way, as an instrument for long-run spending. The control function adds to the probit model the residual from a regression of long-run spending on current income and the other covariates. With the control function, the price effect is $b = 0.52$ $[0.37, 0.68]$, about the same as with long-run spending alone.

Across the other checks, d stays between -0.014 and $+0.031$. Apart from current income, one check gives an unexplained trend whose interval excludes zero. Trimming spending at the 0.5th and 99.5th percentiles gives $d = +0.019$ $[+0.004, +0.033]$, so in that check the spending threshold fell slightly more slowly than the computer price explains.

Fits without survey weights move the change in the spending threshold from 1994 by at most 0.03 (baseline regression) and 0.10 (extended regression, before 2013). A heteroskedastic probit model lets the spread of cutoffs σ differ with the number of interviews and with whether the household is young or a student. The heteroskedastic probit model moves the change in the spending threshold by up to 0.45 in the baseline regression before 2013, where the estimated spread for young or student households hits its bound in 8 years. The heteroskedastic probit model moves the change in the spending threshold by at most 0.06 in the extended regression before 2013, where the household traits already capture young and student households.

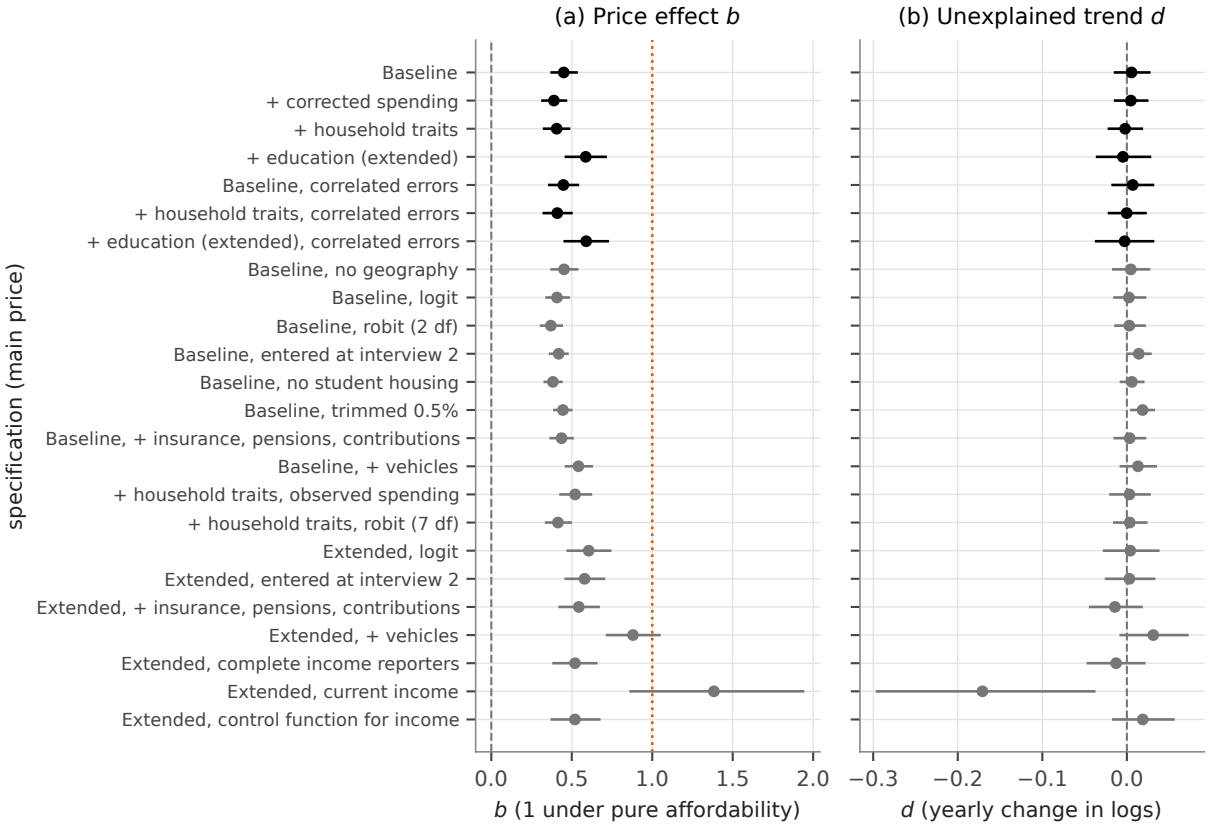
The two-stage method recovers known answers from fake data. We simulated fake computer ownership data with known a, b, d, and ω on the actual households and survey weights, and we fitted the same two stages to the fake data. In fake data with a price effect and no unexplained trend, the method recovers $b = 0.39$ $[0.33, 0.45]$ against a true value of 0.43. In fake data with half that price effect and an unexplained trend, the method recovers $b = 0.24$ $[0.19, 0.29]$ against a true value of 0.22, and $d = -0.040$ $[-0.052, -0.028]$ against a true value of -0.047. An unexplained trend of that size would have shown up in the real data. The replicate standard errors are about right. Across the 20 spending thresholds, the largest gap between an estimated and a true spending threshold, divided by its replicate standard error, is 1.84 in the first fake-data scenario and 2.05 in the second.

One joint model of all households gives the same answer. Fitting the extended regression's two stages at once, as one Bayesian model of all 178,891 CE households of 1994–2013, gives $b = 0.60$ $[0.45, 0.74]$ and $d = -0.003$ $[-0.038, 0.029]$, against 0.59 $[0.46, 0.72]$ and -0.005 $[-0.037, 0.029]$ from the two stages. The joint model's intervals do not allow for the survey design, so the comparison of the joint model with the two-stage fit is a comparison of point estimates (Online Appendix G).

Table A2. Checks under the main computer price: posterior medians with 90% intervals. The second-to-last column is the observed change in the log spending threshold from 1994 to 2012.

Check	b	d	ω	$\hat{\tau}_{2012} - \hat{\tau}_{1994}$	max $\hat{R}$
Baseline, no geography	0.45 [0.37, 0.54]	+0.005 [−0.018, +0.028]	0.027	−1.65	1.003
Baseline, logit	0.41 [0.34, 0.49]	+0.002 [−0.016, +0.023]	0.029	−1.54	1.002
Baseline, robit (2 df)	0.37 [0.30, 0.45]	+0.003 [−0.015, +0.023]	0.029	−1.39	1.002
Baseline, households entering at interview 2	0.42 [0.36, 0.48]	+0.014 [−0.001, +0.029]	0.025	−1.42	1.001
Baseline, no student housing	0.38 [0.32, 0.44]	+0.006 [−0.008, +0.021]	0.022	−1.44	1.003
Baseline, spending trimmed at 0.5% and 99.5%	0.44 [0.38, 0.50]	+0.019 [+0.004, +0.033]	0.027	−1.46	1.003
Baseline, spending with insurance, pensions, contributions	0.44 [0.36, 0.51]	+0.003 [−0.016, +0.023]	0.020	−1.67	1.003
Baseline, spending with vehicles	0.54 [0.46, 0.63]	+0.013 [−0.009, +0.036]	0.030	−1.89	1.002
+ household traits, observed spending	0.52 [0.42, 0.63]	+0.003 [−0.021, +0.028]	0.047	−1.94	1.001
+ household traits, robit (7 df)	0.41 [0.33, 0.50]	+0.003 [−0.016, +0.024]	0.044	−1.54	1.002
Extended, logit	0.61 [0.47, 0.75]	+0.004 [−0.029, +0.039]	0.063	−2.15	1.002
Extended, households entering at interview 2	0.58 [0.45, 0.71]	+0.003 [−0.026, +0.034]	0.051	−2.09	1.003
Extended, spending with insurance, pensions, contributions	0.54 [0.42, 0.67]	−0.014 [−0.045, +0.019]	0.052	−2.26	1.002
Extended, spending with vehicles	0.88 [0.71, 1.05]	+0.031 [−0.009, +0.073]	0.071	−2.82	1.002
Extended, complete income reporters	0.52 [0.38, 0.66]	−0.013 [−0.048, +0.022]	0.063	−2.18	1.002
Extended, current income	1.38 [0.86, 1.95]	−0.171 [−0.297, −0.037]	0.259	−7.98	1.005
Extended, control function for income	0.52 [0.37, 0.68]	+0.019 [−0.018, +0.056]	0.073	−1.68	1.003

Figure A2. Under most choices of sample, spending measure, and link function, the price effect stays between 0.37 and 0.61, and the unexplained trend stays near zero



The price effect b and the unexplained trend d in every specification, main computer price: posterior medians and 90% intervals. Black: the four CE regressions of 1994–2013, from the baseline to the extended regression, with independent and with correlated sampling errors. Grey: checks. Pure affordability is $b = 1$ (dotted vertical line in panel a) and $d = 0$. Dashed lines mark zero.

A.3 The education shift under other specifications

The college shift stays large under every other specification of the extended regression. Table A3 gives the education shifts in the log spending threshold, averaged over 1994–2012. With corrected long-run spending, a college degree lowers the log spending threshold by 1.04 $[0.98, 1.11]$ relative to a high-school education, some college lowers the log spending threshold by 0.67, and less than high school raises the log spending threshold by 0.51. The college shift is 0.79 to 1.14 in every year from 1994 to 2012 except 2003 and 2006 (1.55 and 1.45), the two years with the flattest spending slope. With current income in place of long-run spending, the education shifts are four times larger (college: 4.30), because current income measures long-run resources poorly and education stands in for the missing resources. With the control function of Appendix A, which uses current income as an instrument for long-run spending, the college shift shrinks to 0.77 $[0.69, 0.86]$, a spending ratio of 2.2. In the raw CE data, with survey weights, the computer ownership rate of households headed by a college graduate rose from 44.5% in 1994 to 93.8% in 2012, and that of households headed by a high-school graduate rose from 14.7% to 70.9% (`education_raw_gap.py`).

Table A3. Education shifts in the log spending threshold, θ_e, averaged over 1994–2012, with 90% intervals from the replicates of the mean. A positive shift means that households with that education own a computer at lower spending than households headed by a high-school graduate. A shift of 1.04 means that a household with that education was as likely to own a computer as a household headed by a high-school graduate that spent 2.8 times as much. The current income, complete income reporters, and control function rows use the 144,733 households that report complete, positive income. The logit model's $\hat{\beta}_t$ is on the logistic scale.

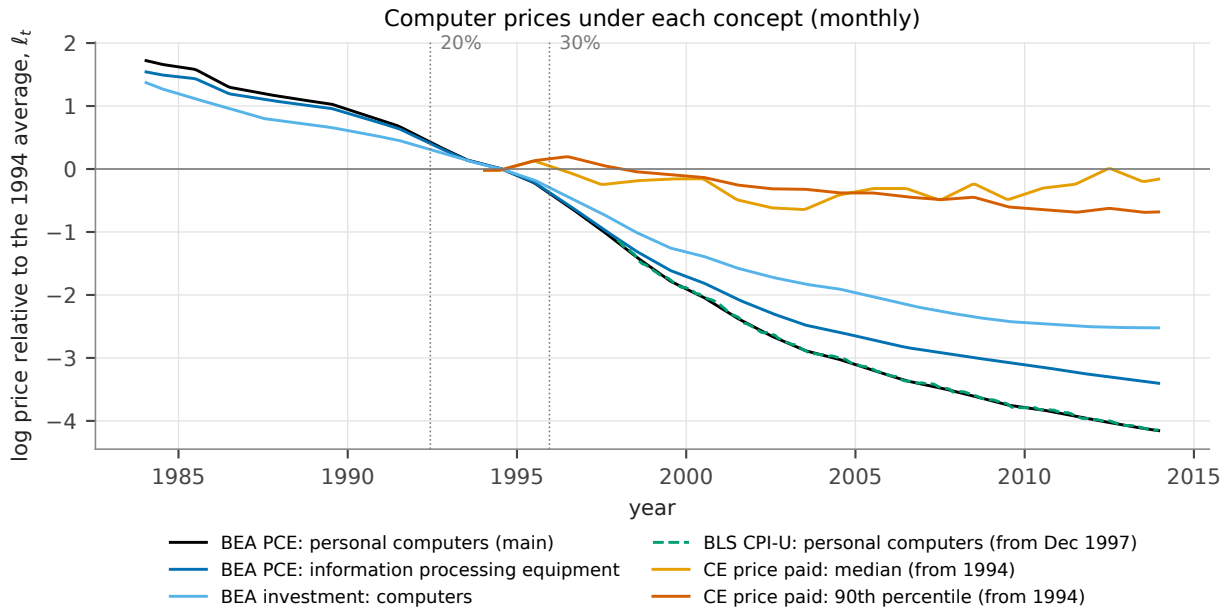
Spending or income measure	Less than high school	Some college	College degree	Mean $\hat{\beta}_t$
Corrected spending, all households	−0.51 [−0.55, −0.47]	+0.67 [+0.62, +0.73]	+1.04 [+0.98, +1.11]	0.70
Corrected spending, logit	−0.50 [−0.54, −0.46]	+0.66 [+0.60, +0.71]	+1.03 [+0.97, +1.09]	1.23
Corrected spending, complete income reporters	−0.49 [−0.53, −0.45]	+0.64 [+0.59, +0.70]	+1.00 [+0.94, +1.07]	0.74
Spending with a control function for income	−0.38 [−0.42, −0.33]	+0.52 [+0.46, +0.58]	+0.77 [+0.69, +0.86]	0.89
Current income	−2.06 [−2.24, −1.87]	+2.60 [+2.34, +2.85]	+4.30 [+3.93, +4.67]	0.21

A.4 Other computer prices

Figure A3 plots every computer price concept, monthly from 1984 to 2013, relative to its 1994 average. Table A4 repeats the second stage under every computer price concept. The two other BEA indexes give similar results, with b between 0.39 and 0.79 and unexplained trends between −0.030 and −0.009 a year. The unexplained trends under the two other BEA indexes are all negative, because these two indexes fell less than the main index and so leave more of the fall in the spending threshold to the unexplained trend.

The CPI-U for personal computers gives $b = 0.63$ in the baseline regression and 0.93 in the extended regression, with a positive unexplained trend (extended regression: 0.056 $[0.005, 0.106]$). The difference from the main index reflects the shorter period that the CPI-U covers, 1999 to 2013. The CPI-U and the main index both fall by 0.18 a year in logs from 1999 to 2012, and over those years the computer price falls at an almost constant rate, so the CPI-U fits cannot separate b from d (posterior correlation 0.98).

Figure A3. From 1994 to 2012 the quality-adjusted computer price fell to 1/56 of its 1994 level, while the median dollar price that households paid for a computer fell only 7%



Log computer price under each price concept, monthly from 1984 to 2013, relative to the 1994 average. A fall of 1 in the log price is a fall to 37% of the earlier price, and a fall of 4 is a fall to 1.8%. The BLS consumer price index for all urban consumers (CPI-U) for personal computers starts in December 1997 and is anchored to the main index's 1999 level. The price paid is the CE's weighted median and 90th percentile of reported purchase prices for computers.

Table A4. Second stage under every computer price concept: posterior medians with 90% intervals. $\Delta\ell$ is the change in the log computer price from the first year used to 2012. The CPI-U starts in 1999.

Specification	Years	$\Delta\ell$ to 2012	b	d	ω
BEA PCE, personal computers (main)					
Baseline	1994–2013	−4.02	0.45 [0.37, 0.54]	+0.005 [−0.016, +0.028]	0.029
+ corrected spending	1994–2013	−4.02	0.39 [0.31, 0.47]	+0.005 [−0.015, +0.026]	0.034
+ household traits	1994–2013	−4.02	0.41 [0.32, 0.49]	−0.002 [−0.023, +0.019]	0.043
+ education (extended)	1994–2013	−4.02	0.59 [0.46, 0.72]	−0.005 [−0.037, +0.029]	0.062
BEA PCE, information processing equipment					
Baseline	1994–2013	−3.30	0.45 [0.36, 0.56]	−0.014 [−0.034, +0.006]	0.045
+ corrected spending	1994–2013	−3.30	0.39 [0.30, 0.49]	−0.012 [−0.032, +0.007]	0.047
+ household traits	1994–2013	−3.30	0.40 [0.30, 0.51]	−0.021 [−0.041, −0.001]	0.055
+ education (extended)	1994–2013	−3.30	0.59 [0.43, 0.76]	−0.030 [−0.063, +0.004]	0.079
BEA investment, computers					
Baseline	1994–2013	−2.53	0.62 [0.47, 0.77]	−0.011 [−0.034, +0.014]	0.058
+ corrected spending	1994–2013	−2.53	0.53 [0.39, 0.67]	−0.009 [−0.031, +0.012]	0.056
+ household traits	1994–2013	−2.53	0.54 [0.37, 0.70]	−0.019 [−0.044, +0.006]	0.069
+ education (extended)	1994–2013	−2.53	0.79 [0.55, 1.04]	−0.027 [−0.063, +0.012]	0.096
BLS CPI-U, personal computers					
Baseline	1999–2013	−2.36	0.63 [0.44, 0.79]	+0.038 [+0.001, +0.070]	0.023
+ corrected spending	1999–2013	−2.36	0.56 [0.39, 0.70]	+0.034 [+0.003, +0.064]	0.020
+ household traits	1999–2013	−2.36	0.61 [0.42, 0.77]	+0.034 [−0.003, +0.065]	0.033
+ education (extended)	1999–2013	−2.36	0.93 [0.67, 1.18]	+0.056 [+0.005, +0.106]	0.043
CE price paid, median					
Baseline	1994–2013	−0.08	0.75 [0.55, 0.96]	−0.087 [−0.095, −0.079]	0.066
+ corrected spending	1994–2013	−0.08	0.65 [0.47, 0.84]	−0.075 [−0.082, −0.068]	0.062
+ household traits	1994–2013	−0.08	0.67 [0.48, 0.86]	−0.084 [−0.091, −0.078]	0.063
+ education (extended)	1994–2013	−0.08	0.96 [0.67, 1.27]	−0.123 [−0.134, −0.111]	0.094
CE price paid, 90th percentile					
Baseline	1994–2013	−0.64	0.77 [−0.07, 1.59]	−0.063 [−0.103, −0.025]	0.140
+ corrected spending	1994–2013	−0.64	0.68 [−0.09, 1.41]	−0.054 [−0.090, −0.019]	0.123
+ household traits	1994–2013	−0.64	0.80 [−0.02, 1.56]	−0.059 [−0.097, −0.022]	0.124
+ education (extended)	1994–2013	−0.64	1.08 [0.09, 2.06]	−0.089 [−0.136, −0.041]	0.172

B Further results

B.1 The main estimates in plain units

Table B1 lists the main estimates in plain units, in the order of the paper's three key results.

B.2 Pure affordability predicts a faster spread of computers than occurred

Against the main computer price, pure affordability overpredicts both the fall in the spending threshold and the rise in the computer ownership rate. The gap from pure affordability, L_t (Online Appendix D.3), reaches 1.79 to 2.57 by 2012 (extended regression: 1.79, SE 0.17). The log spending threshold fell 1.8 to 2.6 less than the log computer price, so by 2012 the spending threshold was 6 to 13 times the level that pure affordability predicts. For the computer ownership rate, the fitted 1994 probit model, applied to 2012 households at the 2012 computer price, predicts a rise 1.29 to 1.34 times the actual rise (S_{2012}). That prediction implies that nearly every household owned a computer by 2012, against the actual 81%. Spending growth alone accounts for 20–25% of the actual rise (S_{2012}^{spend} of 0.20 to 0.25), in current dollars, which count general inflation as spending growth. Section 5.1 splits the rise between the computer price and household spending at the estimated b and in constant dollars.

B.3 The CPS and the test for a switch at 20–30% ownership

The CPS extension gives no reliable evidence that the unexplained trend changed when the computer ownership rate passed 20% or 30%. With both surveys, which together cover 1984 to 2013, the model with a switch at a computer ownership rate of 30%, in late 1995, best predicts the spending thresholds left out of the fit. The 30% switch model is ahead of the one-trend model by 8.7 (SE 4.5) in expected log predictive density (elpd, a measure of how well a model predicts left-out data, where higher is better). Under the 30% switch model the unexplained trend changes by -0.096 $[-0.132, -0.060]$ a year at the switch date, with $b = 0.46$ $[0.27, 0.67]$ (Figure B1).

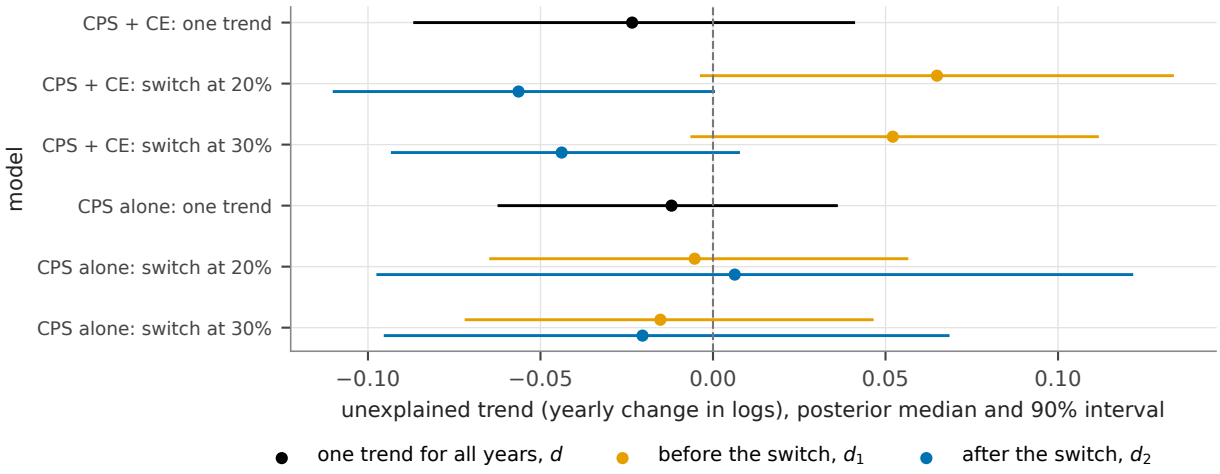
We do not believe that the switch at 30% is real. The switch date falls where the data change from the CPS, which supplies most of the spending thresholds before the switch date, to the CE, which supplies most of the spending thresholds after the switch date. The CPS imputes spending from bracketed income, and in the years that both surveys cover, the imputation compresses the path of the spending threshold by about a third (Online Appendix F). Within the CPS alone, the one survey that measures computer ownership consistently across both switch dates, the model with one unexplained trend predicts best, ahead of the 30% switch model by 2.1 (SE 0.6). Within the CPS, the change in the unexplained trend is -0.005 $[-0.038, 0.037]$ at 30% and 0.013 $[-0.050, 0.079]$ at 20%.

Only the CPS supplements of October 1984 and October 1989 measure computer ownership below 20%, and these two supplements have the largest standard errors, so the surveys contain little information about a switch at 20%. The early phase of the spread of computers is where the profit switch hypothesis is most distinctive, and the surveys have the fewest observations in that early phase.

Table B1. Main estimates in plain units, CE households, 1994–2013, with changes measured from 1994 to 2012. The price effect b and the unexplained trend d come from the second stage under the main computer price, the Bureau of Economic Analysis price index for personal computers. The other model rows use the baseline regression, except the education row, which uses the extended regression (Section 4). The gap row and the price row use the raw survey data and the price indexes.

Quantity	Estimate [90% interval]	What it means
Price effect b, baseline and extended regressions, 1994–2013	0.45 [0.37, 0.54] and 0.59 [0.46, 0.72]	A 10% fall in the quality-adjusted computer price lowered the spending threshold by 4.6% in the baseline regression and by 6.0% in the extended regression. Pure affordability predicts a 10% fall in the spending threshold.
Unexplained trend d, baseline and extended regressions, 1994–2013	$+0.005$ $[-0.016, 0.028]$ and -0.005 $[-0.037, 0.029]$ a year	Once the computer price is taken into account, the spending threshold shows no other steady change. The intervals rule out unexplained trends larger than about 4% a year in either direction.
Share of the 1994–2012 rise in the computer ownership rate attributed to the falling computer price, constant dollars	94% (93–96% over the interval for b)	Of the 56 percentage points of the rise that computer prices and household spending explain, the falling computer price accounts for 53 points and rising household spending for 3 points.
Fall in the dollar price that households paid for a computer, 1994–2012	7% at the median and 47% at the 90th percentile (0.076 and 0.638 in logs), against 98% (4.02 in logs) for the quality-adjusted computer price	Better computers account for 84–98% of the fall in the log quality-adjusted computer price, and lower dollar prices account for the rest.
Gap in computer ownership between the richest and poorest fifths of households, ranked by reported income	39.3 points in 1994; 53.8 in 2003; 36.6 in 2012	Every household faced the same price. The gap in computer ownership between rich and poor households widened from 1994 to 2003, while most households close to buying a computer had high incomes, and narrowed from 2003 to 2012, when most households close to buying had low incomes.
Rise in the computer ownership rate from a 10% cut in the computer price	1.10 points [0.89, 1.32] in 1994; 1.37 in 1999–2000; 0.94 [0.77, 1.12] in 2012	The rise was largest when the spending threshold was near the spending of the median household. The intervals allow for uncertainty in b only.
College degree: shift in the log spending threshold, extended regression, average over 1994–2012	1.04 [0.98, 1.11]	A household headed by a college graduate was as likely to own a computer as an otherwise similar household headed by a high-school graduate that spent 2.8 times as much.

Figure B1. The unexplained trend appears to change at 30% ownership when the CPS and the CE are combined, but within the CPS alone the unexplained trend shows no change



Unexplained trend before and after the switch date, main computer price: posterior medians and 90% intervals. CPS + CE: the switch test with a CPS offset, 1984–2013. CPS alone: the check on the CPS spending thresholds, October 1984 to October 2003.